WORKING DRAFT — NOT FOR SUBMISSION

Planning What You Cannot Count: An Open-Data Framework for Bus Route Rationalisation and Fleet Sizing under Demand-Data Scarcity

Target journal: Transport Policy (Elsevier)

Assembled 2026-10-08 · repository revision **d25a39c**

What this document is. Every section file in `paper/sections/` rendered in full, in order, with nothing removed. It exists so that co-authors can read their own text in context and act on the open questions in it.

What has been done to co-author prose: nothing. Text written by a co-author is transcribed verbatim. Where it conflicts with another part of the manuscript, the conflict is raised in a tinted callout and *proposed* replacement wording is given — the prose itself is left alone. There is exactly one exception, a section heading, and it is disclosed at the point of change (FLAG 3-A).

What the callouts mean. Orange = an editorial decision a co-author must take. Red = a blocker that can sink the submission if unresolved. Blue = provenance or neutral notes. The companion briefing `PENDING_DECISIONS.pdf` collects every one of them into numbered decisions with options and consequences.

Author list, affiliations, corresponding author and CRediT roles are deliberately absent — they are pending (Decision 8).

Contents, ownership and current state

#	Section	Owners	Words (target)	Status
00	Abstract	Prashant, Ankit, Sharvesh	228 (250)	Complete draft on locked numbers
01	1. Introduction	Misti, Avny	2,343 (900)	Co-author prose preserved verbatim + editorial flags
02	2. Literature review and research gap	Sharvesh, Ankit	964 (900)	Rewritten on the executed review (D5 = A); Table 1 in supplement
03	3. Study area and data	Krishna	1,242 (700)	Drafted for co-author review; ethics paragraph blank
04	4. Methodology	Prashant (sole owner)	2,239 (1,700)	Complete draft; Eqs 1-14 + Algorithm 1
05	5. Results	Prashant, Misti	2,412 (1,800)	All subsections drafted; §5.12 provisional (D8)
06	6. Validation and robustness	Avny, Krishnan, Prashant	1,041 (650)	V1, V2, V4, V5, V6 executed; V3 forward work
07	7. Discussion and policy implications	Ankit, Avny, Misti	2,045 (1,000)	Drafted for co-author review + co-author block verbatim

#	Section	Owners	Words (target)	Status
08	8. Conclusions	Ankit, Prashant, Sharvesh	475 (350)	Complete draft
	Total body words		**12,989**	target ≤ 8,000 (Transport Policy)

Word counts are measured from the section files at build time and exclude headings, tables, code blocks and every editorial callout — i.e. they count only prose that would survive into a submitted manuscript. They are therefore lower than the raw file length.

Abstract

Section owners: Prashant, Ankit, Sharvesh. **Convention:** draft last. **Status (2026-09-30):** rewritten on the executed modules, including the a09 fleet interval. Every value cites `[CL-xx]`. Kashmir-Division scope; the legacy "342 permits → 207 routes / ~1,009 fleet / Srinagar-metropolitan" framing is **superseded and barred**. ~230 words.

Most Indian cities that need bus-network reform lack the origin–destination and ridership data that route-optimisation methods assume. We present an open-data, demand-free framework for bus route rationalisation and fleet sizing and apply it to Kashmir Division, India (6,584,762 residents) `[CL-11]`, using a gridded population surface, OpenStreetMap, a routing engine and the digitised permit register. The register's 614 permits describe only 157 corridors, so the plan's apparent 71 % "route reduction" is 99.4 % corridor retention plus a change of unit `[CL-01, CL-02, CL-07, CL-08]`. Measuring catchments on the pedestrian network rather than as straight-line buffers lowers population served by a median of **37.4 % per route** and division coverage from **35.5 % to 24.2 %** `[CL-26, CL-28]`; only 10.4 % of residents live near a 15-minute service `[CL-42]`. Against 43,809 driver-GPS runs, modelled run times are too fast and a per-kilometre cap, not demand, sets cycle time on 169 of 186 routes `[CL-18, CL-31]`. Joint uncertainty analysis therefore reports the fleet as a range: 989–1,058 buses as specified, but **1,130–1,266 at observed pace**, against a published 1,011; the service hierarchy is robust (97.8 % tier agreement) `[CL-56, CL-57]`. Thirty per cent of the fleet, bought in order of new residents reached per bus, delivers 92 % of the plan's coverage `[CL-61]`. The plan is decision-robust in its hierarchy, not demand-validated.

Keywords: bus route rationalisation; demand-data scarcity; open data; network walk catchments; fleet sizing; uncertainty analysis; Kashmir

[PRESERVED CO-AUTHOR PROSE IN THIS SECTION] Parts of this section are transcribed verbatim from the co-author attachment and have not been edited, condensed or reordered.

1. Introduction

Provenance and editorial status. Sections 1.1–1.6 below are **co-author prose, transcribed verbatim** from the author attachment "*Copy of Paper Route Rationale_AK.pdf*" (owners: Misti and Avny; drafted text recovered via `paper/_reflow_ak.py`). **Nothing in that prose has been rewritten, condensed, or deleted.** The only mechanical interventions are (a) restoring section headings that the flat PDF extraction lost, (b) rejoining two sentences that a page break had split mid-number, and (c) adding clearly delimited `> **[EDITORIAL FLAG …]**` blocks. Every flag raises a question for the co-authors; none resolves one. Target length for this section is 1,200 words (attachment). Current verbatim body: ~1,280 words.

1.1 Urban bus transport under stress in Indian cities

India is facing a structural and widening shortage for urban public buses. In estimation from 2026 to 2047, the country's urban population is to grow from 490 million to about 763 million; this adds more than 270 million residents who will have dependencies for public transport (CSE & CITIES Forum 2026). In contrast to this demographic situation and demand, the municipal bus fleets continue to shrink relatively. The current national bus availability ranges roughly from 0.13 to 0.14 buses per 1000 people, this makes national provision nearly 70% below the benchmark of 0.44 set by Ministry of Housing and Urban Affairs (MoHUA) as per CSE & CITIES Forum, 2026; MoHUA, 2014; Pucher et al., 2007. This shortage issue hits secondary cities most. Over 400 Class-I cities with population count ranging from 100,000 to 1 Million have no organised municipal bus network at all (CSE-CITIES Forum, 2026) while at the same time base digital transit data is almost non existing; less than 15 indian transit agencies publish open General Transit Feed Specification (GTFS) feeds. On ground, this gap is covered with private minibuses and other vehicles. Operating through this ad-hoc, permit by permit allocation without route integration or systematic timetable leaves the urban cities with fragmented and unreliable service to public.

[EDITORIAL FLAG 1.1-a — citation verification] Five quantitative claims in this paragraph each need a page-level citation before submission: the 490→763 million urban projection; the 0.13–0.14 buses/1,000 figure; the MoHUA 0.44 benchmark; the ">400 Class-I cities with no organised bus network" count; and the "<15 agencies publish GTFS" count. `paper/references.bib` carries the CSE & CITIES Forum (2026), MoHUA (2014) and Pucher et al. (2007) entries; the specific page/table for each number is still needed. This is the highest-yield paragraph in the paper for a reviewer to fact-check, because every number in it is externally verifiable.

[EDITORIAL FLAG 1.1-b — in-text citation format] "as per CSE & CITIES Forum, 2026; MoHUA, 2014; Pucher et al., 2007" should be parenthetical *(CSE & CITIES Forum, 2026; MoHUA, 2014; Pucher et al., 2007)* in Elsevier/APA style. Also note "CSE & CITIES Forum 2026" and "CSE-CITIES Forum, 2026" appear as two different strings for the same source — one form must be chosen.

1.2 Route rationalisation as a low-cost, high-leverage intervention

Urban transport policies often emphasize for capital-intensive, mass transit infrastructure to address the growing demand. Though the urban rail metro systems provide substantial capacity along high density corridors in primary metropolitan areas, they require significant capital for enabling route to over US$ 40 Million per km, compared to US$ 2.5 Million per km when its Bus Rapid Transit (BRT) (Badami & Haider, 2007). In secondary and Tier-II urban regions, taking mutli-year timeline projects and heavy infrastructure costs creates need for careful fiscal planning along with catering needs for

immediate transit needs. Instead, comprehensive bus route rationalisation along with network redesign can make an exceptional cost effective and efficient solution that costs under US$0.05 Million per km (Badami & Haider, 2007). Rather than relying on heavy civil construction, this restructuring optimizes for existing ground infrastructure and operational vehicle assets. By merging overlapping services on busy central spines and deploying the spare assets to cover underserved peripheral corridors, route rationalisation delivers rapid improvements for better public access. This is even positively practical and cost efficient for municipal and transport departments that seek solution for delivery of time effective results on ground for people.

[EDITORIAL FLAG 1.2-a — page-break repair, disclosed] The PDF extraction split "US$ 40 Million per km, compared to US$ / 2.5 Million per km" across a page boundary. The two fragments have been rejoined as "US$ 2.5 Million per km". No wording was altered. If the intended figure was different, correct here.

[EDITORIAL FLAG 1.2-b — cost figures need a year and a base] The three capital-cost figures (US$40M/km metro, US$2.5M/km BRT, US$0.05M/km rationalisation) are attributed to Badami & Haider (2007). A 2007 dollar figure quoted in a 2026 paper needs either the base year stated explicitly or inflation adjustment. A reviewer will ask. Recommend: "(in 2007 US$; Badami & Haider, 2007)".

1.3 The demand-data precondition and why it binds

The academic literature in Transit Network Desgin and Frequency Setting (TNDFS) provides extensive mathematical frameworks to optimize bus routes and their operating schedules (Ceder, 2007; Guihaire & Hao, 2008; Ibarra-Rojas et al., 2015). However, these methodologies have the fundamental precondition: the availability of comprehensive, empirical passenger demand data, such like household origin-destination (OD) surveys, automated fare collection (AFC) transactions or automatic passenger counts (APC). In real practise, this demand creates bottleneck across four core stages of public transit planning:

1 **Objective function formation:** Classical network optimization models minimize generalized passenger travel disutility like in-vehicle time, waiting time and transfer penalties while balancing against operator cost.

1 **Frequency and Headway setting:** Standard operational methods calculate route frequency (*f_k*) as a function of peak passenger volume on the heaviest loaded link (*Q_{k,max}*):

<!-- FORMULA PLACEHOLDER — see EDITORIAL FLAG 1.3-a -->

Where *C_{veh}* is rated vehicle capacity and *LF_{target}* is the target load factor. If peak passenger volume is unobserved, frequency cannot be derived.

1 **Route Pruning:** Standard algorithms identify and eliminate redundant routes based on passenger-kilometre revenue or observed boarding numbers. Without electronic ticketing data, authorities have no reliable way to separate essential trunk routes from redundant parallel ones.

1 **Model Validation:** Conventional planning pipeline validate their results by back-casting simulated passenger flows against historical counts.

For mid-sized and secondary cities, this creates an operational deadlock. Local authorities cannot run standard network optimization without passenger demand data and also lack the infrastructure or survey budget to collect the needful data. This seeks rethinking of planning model. As shown in **Figure 1**, instead of waiting for unavailable demand data, cities can adopt a supply-side approach which builds an implementable network using open spatial data and clear, policy-driven service standards.

[EDITORIAL FLAG 1.3-a — missing equation] The PDF carries a `<formula>` placeholder here; the equation itself did not survive extraction. The standard form implied by the surrounding text is *f_k = Q_{k,max} / (C_{veh} · LF_{target})*. **This has NOT been inserted into the body** — the co-authors must supply and confirm their intended equation, since it is their prose and their citation. It is worth numbering it as Eq. (0) or moving it to §2, because §4 already owns Equations 1–14 and a stray unnumbered formula in §1 will confuse the cross-reference scheme.

[EDITORIAL FLAG 1.3-b — this is the load-bearing subsection] The attachment's own guidance is that "§1.3 is the section that earns the paper … If the reader does not feel the constraint bite here, nothing later lands." The four-stage breakdown does this well. One strengthening suggestion (co-authors' call): each of the four stages could name the *specific* data object it requires — stage 1 an OD matrix, stage 2 link loads, stage 3 boarding counts, stage 4 a historical count series — so the reader can tick off exactly what Kashmir lacks.

1.4 Data scarcity in Tier-II, permit-based networks

In secondary Indian cities, route-level permits stand as base for evolving the public transit network through history, issued by Regional Transport Offices (RTOs) and State Transport Authorities (STAs) in response to localized mobility demands. Over time, this point to point allocation has naturally concentrated them highly along primary commercial and healthcare arterials, while peripherals and developing peri-urban links suffer in frequency and service. In addition, the public transit delivery involves multiple stakeholders including RTOs managing private stage carriages, municipal Special Purpose Vehicles (eg: Smart City Ltd) introducing modern electric buses, and State Road Transport Corporations (SRTCs) — each operating under distinct mandates without an integrated digital scheduling platform or system.

The Kashmir Valley illustrates this operational reality at a regional scale. It comprises of 10-district urban-rural basin of 15,948 sq.km with 6.58 million residents. The valley in itself has unique physical constraints for mobility network, including nine critical Jhelum River Bridge crossing in Srinagar, major lakes barriers (Dal, Nigeen and Anchar), and sub-zero winter freezes during Chillai Kalan. The existing network comprises of 640+ legacy permit routes, with heavy overlap along primary civic arterials — such as 38 to 42 active permits serving the Soura and Lal Ded hospital corridors. The network must also consider and support the significant gender-specific transit dependency (64.5% female ridership recorded via CHALO electronic ticketing) and an annual influx of 2.5 million tourists, all of this with lack of a unified open GTFS feed.

[EDITORIAL FLAG 1.4-a — this paragraph is ALREADY on the locked framing. Good.] §1.4 uses the correct Kashmir-Division scope (10 districts, 15,948 km², 6.58 million, 640+ permits, 38–42 permits on the Lal Ded corridor). This matches `STATUS.md §0`, `CLAIM_LEDGER.md` (CL-01, CL-05) and F1. **No change needed.** Note for the co-authors: this is the framing the whole paper now uses, and it *contradicts* the Abstract plan, the Highlights and the "Policy Failure" block in the same attachment, which still say "342 permits → 207 routes / ~1,009 buses / Srinagar Municipal Corporation". See `PENDING_DECISIONS.pdf` Decision 3.

[EDITORIAL FLAG 1.4-b — two numbers need a source line] (i) "64.5% female ridership recorded via CHALO electronic ticketing" — this is not currently in the claim ledger and no module produces it. It needs a CHALO source document and a new CL ID, or it must be dropped. (ii) "2.5 million tourists" annually needs a JKTDC / J&K Economic Survey citation with a year. Both are strong, quotable numbers — worth the ten minutes to source them properly.

[EDITORIAL FLAG 1.4-c — attachment guidance not yet followed] The attachment instructs: "*Introduce Srinagar in three sentences at the end of 1.4 … Do not open the paper with Kashmir. Transport Policy wants the problem class first, the city second.*" §1.1–§1.3 correctly keep the problem class first, and the valley enters only in §1.4 — so the guidance **is** satisfied, but the valley paragraph runs to seven sentences rather than three. Co-authors may wish to compress, or may reasonably decide the extra detail earns its place given it now covers a 10-district region rather than one city.

1.5 Research gap, objectives and contributions

Despite the advancements in transit network design, present optimization frameworks offer no validated methodology to restructure fragmented, permit based bus systems in complete absence of empirical demand data, which is the case across most secondary and Tier-II Indian cities. To address this gap, this study develops a supply-side, open-data planning framework that translates road topology, gridded population distributions, and community anchors into a rationalised transit network.

Main objectives:

- **O1 (Spatial Baseline):** Construct an open spatial transit foundation integrating OpenStreetMap road networks, WorldPop demographics, and geo-located Points of Interest (POI) across a regional basin.
- **O2 (Topological Consolidation):** Develop an undirected graph consolidation algorithm to prune overlapping legacy permits into a coherent multi-tier corridor hierarchy.
- **O3 (Supply-Side Scheduling):** Derive policy-anchored headway bands from a Composite Demand Index (CDI) independent of unobserved passenger link flows.
- **O4 (Fleet Dimensioning):** Calculate realistic round-trip cycle times, fleet allocations, and vehicle size classes (light, medium, and heavy passenger vehicles: LPV, MPV, HPV) accounting for physical bridge bottlenecks and dwell penalties.
- **O5 (Validation & Portability):** Benchmark the operational outputs against empirical operational data and demonstrate policy transferability.

Correspondingly, this paper sets five major contributions: (1) mathematical rigorous, supply-side methodology of network design for data scarce regions; (2) a graph-based algorithm to merge overlapping route permits into unified corridors; (3) a novel demographic based and seasonal considerate demand-proxy formulations; (4) a regional-scale deployment across 10 districts in the Kashmir Valley, restructuring 644 legacy permits into 186 coordinated corridors; and (5) an open, reproducible spatial workflow designed for adoption across data-constrained secondary urban regions.

[EDITORIAL FLAG 1.5-a — contribution (4) is exactly right and should be protected] "644 legacy permits into 186 coordinated corridors" matches the locked numbers (CL-05, CL-06). Keep this sentence verbatim; it is the single place in §1 where the headline result is stated, and it is stated correctly.

[EDITORIAL FLAG 1.5-b — O5 wording vs. what §6 can actually claim] O5 says "Benchmark the operational outputs against empirical operational data". §6 as drafted can support this **only** in the supply-side sense (geometry → cycle time → fleet, validated against 43,809 driver-GPS runs; F3, F10–F12). It cannot support a demand-side reading. Recommend tightening O5 to "…against empirical *operational* (supply-side) data", so that no reader takes O5 as a promise of ridership validation. This is a one-word-class change and is left to the co-authors.

[EDITORIAL FLAG 1.5-c — "novel … seasonal considerate demand-proxy" (contribution 3)] The seasonality element is the 1.3× tourist multiplier applied to 8 routes, which F9 diagnoses as sizing 285,914 synthetic headcount. Calling it "novel" invites a reviewer to audit it. Safer framing: describe it as an explicit, parameterised and *sensitivity-tested* seasonal adjustment (it is in the Sobol' set, `a09`), rather than as a novelty. Co-authors' call.

1.6 Organisation of the paper

The remainder of this paper content is as follows. Section 2 reviews the literature on transit network design under data constraints and details the physical and institutional setting of the Kashmir Valley. Section 3 formulates the supply-side methodology, detailing spatial baseline integration, topological permit consolidation, demand proxying, and fleet dimensioning. Section 4 presents the empirical results and sensitivity analyses. Section 5 discusses policy trade-offs, governance challenges, and framework transferability, before Section 6 that concludes with directions for future work.

[EDITORIAL FLAG 1.6-a — □ THE ONE REAL STRUCTURAL CONFLICT IN THE PAPER. Escalated as Decision 1.] This paragraph describes a **six-section** paper (Lit+Setting / Methodology / Results / Discussion / Conclusions). The attachment it sits inside describes an **eight-section** paper, and assigns owners and word budgets to all eight: §1 Introduction (Misti+Avny, 1,200 w) · §2 Literature review and research gap (Sharvesh+Ankit, 1,400 w) · §3 Study area and data (Krishna, 1,100 w) · §4 Methodology (Prashant, 2,300 w) · §5 Results and analysis (Prashant+Misti, 2,400 w) · §6 Validation (Avny+Krishnan+Prashant, 900 w) · §7 Discussion and policy implications (Ankit+Avny+Misti, 1,200 w) · §8 Conclusions (Ankit+Prashant+Sharvesh, 400 w). The repository follows that eight-section scheme.

The prose above has been preserved exactly as written and has NOT been renumbered. But it cannot go to the journal as-is, because under it there is no slot for §6 Validation — the section the attachment itself calls the one that "decides the paper's fate". The eight-section scheme is therefore what the rest of the manuscript is built on.

Replacement text is proposed — not applied — for the co-authors to accept or reject:

"*The remainder of the paper is organised as follows. Section 2 reviews the literature on transit network design and route rationalisation, and quantifies the research gap. Section 3 describes the study area, the data inventory and the data-quality diagnostics. Section 4 formulates the supply-side methodology, detailing spatial baseline integration, topological permit consolidation, demand proxying and fleet dimensioning. Section 5 presents the empirical results. Section 6 sets out the multi-channel validation protocol and what it can and cannot establish. Section 7 discusses policy trade-offs, governance challenges and framework transferability. Section 8 concludes with directions for future work*."

Co-author block — "Policy Failure"

[PROVENANCE] The following is **co-author prose, transcribed verbatim** from the attachment, where it appears immediately after §1.6 under the heading "*Policy Failure*". It is reproduced here unchanged. See the flags beneath it for two placement/scope questions that only the co-authors can settle.

Public transport services operating in Jammu and Kashmir do not fall entirely under a single permitting authority. Bus services operating within cities/districts are permitted by the Regional Transport Authority (RTA), buses originating from other districts and operating within the Srinagar Municipal Area may fall under the permitting jurisdiction of the State Transport Authority (STA), Srinagar and Jammu municipal areas operate electric buses run by the respective Smart City Limited Companies, and finally, the JK Road Transport Corporation (JKRTC) operates intra-state and inter state services. The transport network in J&K therefore includes bus services that are governed by different permitting arrangements and agencies. For the purpose of this study, bus service data for Srinagar (the area falling within the limits of the Srinagar Municipal Corporation (SMC)) was studied. The RTA, issues permits broadly divided into Eastern and Western areas for this purpose. For city bus

services permitted by the RTA, an applicant generally applies for a vehicle and indicates the area of operation, rather than proposing a specific route based on an assessment of passenger demand or the requirements of the wider network. The RTA considers such applications in accordance with the applicable legal and regulatory requirements. The Smart City electric-bus system is a separate planned intervention. There is also no systematic route-rationalisation exercise before permits are allotted. The RTC caters largely to destinations outside the municipal area and its routes were designated decades ago, without having been modified to align with changing demographic and demand patterns. At present, there is an absence of a comprehensive city-wide route plan for public transport and an absence of a single agency that regulates and manages public transport operations.

In Jammu and Kashmir, the procedure for grant of a stage carriage permit is primarily governed by the Motor Vehicles Act, 1988 and the rules made thereunder. Where the proposed Stage carriage service is to be operated within the jurisdiction of a particular Regional Transport Authority (RTA), the application is made before the concerned RTA. Accordingly, in the case of services operating within Srinagar District and falling within the jurisdiction of the RTA Srinagar, the RTA is the competent authority to consider and grant the permit. However, where the proposed service extends beyond the jurisdiction of a single RTA, the matter falls within the competence of the State Transport Authority (STA), particularly where the service originates in one district and terminates in another. The applicant is required to furnish details such as the proposed route in the classified areas (Eastern/Western), vehicle particulars, seating capacity, number of trips, timetable and other prescribed particulars. The competent authority examines the application and considers the proposed service in the context of existing transport arrangements, route feasibility and the requirements of the travelling public. The permit may be granted as applied for, granted with modifications, or refused in accordance with the applicable provisions. Once granted, the permit specifies the area (Eastern/Western) and is subject to conditions relating to the passenger capacity and other operational requirements.

Thus, the permit system in J&K operates through a two-level institutional structure, with the RTA dealing primarily with services confined to its jurisdiction and the STA dealing with services that extend across the jurisdiction of more than one RTA. These permits are permanent in nature, implying they do not have an end validity. Over time, the existing route structure has consequently developed through a combination of existing practices and the demands and representations of the Eastern and Western bus associations. While this allows individual applications to be considered within the existing regulatory framework, it does not provide an opportunity to assess how individual permits respond to travel demand and affect the transport network as a whole. In particular, permit allocation is not systematically preceded by an assessment of route duplication, passenger demand, service gaps, connectivity, fleet requirements or the frequency of services required on different corridors.

This is reflected in the present pattern of public transport services. Some corridors have overlapping services, while other areas have comparatively limited connectivity. Examples of overlapping service patterns include Lal Chowk–Lalbazar and Bagh-i-Shuer–Lal Chowk; Chanapora–Soura and Lal Chowk–Soura; and Malbagh–LD and Soura–LD. In contrast, Lal Chowk–Rangreth and Lal Chowk–PanthaChowk have been identified through official and field-level observations as areas with comparatively inadequate connectivity. These observations suggest that the issue is not simply the number of buses available, but also where those buses operate and how the routes are distributed across the city.

The underlying policy issue is that presently permitting is done as a response to individual permit applications rather than for planning the public transport network as a whole, not accounting for where services are required, where routes overlap, which areas are underserved, or what level of fleet and service frequency is appropriate for individual corridors. Moreover, the permit is valid in perpetuity, hence, it is challenging for the regulator to control, alter and plan public transport routes in the face of dynamic scenarios.

The study examines whether a city-wide bus network can be planned and fleet requirements estimated even where detailed passenger-demand data are not available. It explores the use of

readily available spatial and network information as a starting point for route rationalisation and fleet planning. The intention is not to treat such information as a substitute for actual passenger-demand data, but to provide a practical basis for improving upon the present application driven approach and for developing better data systems over time.

[EDITORIAL FLAG PF-a — placement] This block is the strongest institutional writing in the manuscript and is genuinely distinctive material — very few transit papers explain the *statutory* mechanism that produced the network being rationalised. But structurally it is **§3.3 "Existing supply and institutional structure"** material, not introduction material: it is ~900 words, which alone would push §1 to ~2,200 words against a 1,200-word budget. **Recommendation (not applied): move this block to §3.3, and leave a two-sentence pointer in §1.4.** It has been left in place here so that the co-authors see it exactly where they wrote it.

[EDITORIAL FLAG PF-b — □ scope: "bus service data for Srinagar (… Srinagar Municipal Corporation) was studied". Escalated as Decision 3.] The quantitative plan in this paper covers **all ten districts of Kashmir Division** (614 permits + 30 e-bus routes → 644 rows → 186 active routes; denominator 6,584,762 — CL-01, CL-05, CL-06). The sentence above states the study is confined to the SMC area. Both cannot stand. The likely reconciliation — which the co-authors should confirm rather than have imposed — is that the **institutional and permit-law analysis** is Srinagar-specific (the RTA Srinagar Eastern/Western permit system genuinely is), while the **network plan** is division-wide. If so, one clarifying sentence fixes it, e.g. "*The institutional analysis in this subsection is drawn from the Srinagar Municipal Corporation area, where the RTA's Eastern/Western permit structure is documented; the network plan presented in Sections 4–6 covers all ten districts of Kashmir Division.*" **No such sentence has been inserted.**

[EDITORIAL FLAG PF-c — the named corridors are checkable and worth keeping] Lal Chowk–Lalbazar, Bagh-i-Shuer–Lal Chowk, Chanapora–Soura, Lal Chowk–Soura, Malbagh–LD, Soura–LD (overlapping); Lal Chowk–Rangreth, Lal Chowk–Pantha Chowk (under-served). These are exactly the kind of concrete, falsifiable claim reviewers reward. Module `a10_network_diagnostics` computes per-corridor link duplication from the permit register independently; if its most-duplicated corridors match this list, that is a genuine convergence worth a sentence in §5. If they do **not** match, that disagreement must be reported, not smoothed over.

Figure and table calls for §1

- **Figure 1 — Conceptual framework (two panels).** *(Top)* the conventional demand-driven transit planning pipeline, with data dependencies (OD matrices, AFC feeds, APC counts) drawn as locks that prevent optimisation in data-scarce settings. *(Bottom)* the proposed supply-side open-data pipeline, showing OpenStreetMap street topology, WorldPop gridded demographics and open Points of Interest feeding directly into topological consolidation, policy-anchored headway bands and cycle-time fleet dimensioning. **Status: not yet generated** (`analysis/fig_generate_all.py` is a stub). Caption above is the co-authors' own, transcribed verbatim.
- **Statistics the attachment requires for §1, each needing a verified citation.** (1) buses per 1,000 population, India against comparators; (2) number of Indian cities publishing GTFS; (3) share of Class-I cities with an organised bus service; (4) capital cost per km, metro vs BRT vs network redesign; (5) study area population, vehicle registrations, tourist arrivals. Items 1–4 appear in §1.1–§1.2 above and are covered by FLAG 1.1-a and FLAG 1.2-b. **Item 5 is not yet written**: population is established (6,584,762 — CL-11), but *vehicle registrations* and *tourist arrivals* have no source in the repository and no module produces them. Either a co-author supplies citable figures (J&K Economic Survey; JKTDC arrivals series) or item 5 is reduced to population alone.

2. Literature review and research gap

Section owners: Sharvesh, Ankit. **Target:** ~900 words (D26). **Status (2026-10-04):** rewritten on an executed review. Decision **D5 = Option A**: the search was run (OpenAlex and Scopus, 30 September – 1 October 2026), 60 studies were coded (46 from full text), and the §2.8 cross-tabulation is computed, not asserted. Protocol, query log, screening counts, coding rubric, novelty check and the full Table 1 are in `paper/literature_review/` (Supplementary Material S1). Every citation resolves in `references.bib` and was checked against Crossref. Flags 2-A to 2-D are closed; open points are listed at the end.

2.1 Review protocol

We searched Scopus (title, abstract, keywords) and OpenAlex (title, abstract) for English-language articles, conference papers and reviews from 2000 to 2026. Thirteen query blocks combined planning terms (network design, rationalisation, frequency setting, fleet size), data terms (origin–destination, smart card, data scarcity, open data, accessibility, equity) and context terms (India, Global South, paratransit, permits); six further Scopus queries targeted work resembling our own pipeline. The searches returned 5,488 unique records. A keyword pre-filter left 2,518 for title-and-abstract screening against written criteria (a first pass assisted by a large language model, then checked by the authors), and 873 were eligible. We coded a purposive, theme-stratified sample of 57 of these, plus three seminal earlier papers found by citation chaining, on data intensity, demand input, equity treatment, validation and planning output (Table 1; Supplementary Material S1). Forty-six were coded from full text.

2.2 Network design and route rationalisation

The transit network design and frequency-setting problem has been reviewed repeatedly (Guihaire & Hao, 2008; Kepaptsoglou & Karlaftis, 2009; Ibarra-Rojas et al., 2015; Durán-Micco & Vansteenwegen, 2022), and its operational fundamentals are textbook material (Vuchic, 2005; Ceder, 2016). The object optimised is a route set with frequencies and the objective is defined over passenger flows, so an origin–destination (OD) matrix is an input. Ten of our 60 coded studies are demonstrated only on benchmark or synthetic networks with a given OD matrix, most often the Mandl network (Mandl, 1980; Arbex & da Cunha, 2015; Yoo et al., 2023). Real-city route-and-frequency plans exist, for Rome (Cipriani et al., 2012), Tin Shui Wai (Szeto & Wu, 2011; Szeto & Jiang, 2014), Miami-Dade (Zhao & Zeng, 2008) and Bangalore (Verma et al., 2017), but each starts from a demand matrix.

Restructuring an inherited network is studied less often. Overlapping Istanbul routes have been altered inside an assignment model with a 452-zone OD matrix (Ergun & Kesten, 2011), and 14 overlapping Bangkok routes merged using on-board counts (Hoonsiri et al., 2020). Duplication can be measured from schedule data alone, but that work stops at diagnosis (Barzegari et al., 2026).

2.3 Demand estimation and its data requirements

Where demand is not surveyed it is inferred, and the inference consumes passive data: smart-card and vehicle-location records (Barry et al., 2009; Wang et al., 2011; Alsger et al., 2015), fare cards with GPS (Farzin, 2008), mobile-phone traces (Pinelli et al., 2016; Mittal & Ukkusuri, 2025) or taxi GPS (Chen et al., 2014). Even an attempt to predict passenger flows from network properties alone was trained on fare-collection data (Luo et al., 2020). These sources presuppose electronic ticketing or operator instrumentation, which permit-based minibus systems do not have.

2.4 Demand-independent and accessibility-based approaches

A second tradition plans against potential rather than realised travel: need indices built from census variables (Yao, 2007), accessibility surfaces (Tribby & Zandbergen, 2012; Karner, 2018), and

transit-desert detection that uses no demand data (Carmody et al., 2025). One recent study states the premise plainly, assuming demand proportional to population (Liu & Rosenberg, 2026).

Two results from this tradition bear directly on our method, and we claim neither. First, straight-line buffers overstate catchment population relative to network distance (Gutierrez & Garcia-Palomares, 2008; Biba et al., 2010; El-Geneidy et al., 2014), a bias also shown for Indian cities (Yenisetty & Bahadure, 2020); §4 applies this correction and reports its size for our network. Second, headways can be set without ridership: frequencies on existing lines in Palma were adjusted from a socio-demographic need index (Ruiz et al., 2017). These studies evaluate networks or set a single service attribute. None carries the logic through to a fleet.

2.5 Open data and planning under scarcity

Open inputs are now adequate for such work, with caveats. Gridded population is modelled, not observed (Tatem, 2017), and OpenStreetMap road completeness, although above 80 % globally, is uneven (Barrington-Leigh & Millard-Ball, 2017). Schedule feeds support comparable accessibility measurement (Bok & Kwon, 2016; Wessel & Farber, 2019), and field mapping has produced network data for informal systems in Nairobi, Maputo and Bogotá (Williams et al., 2015; Klopp & Cavoli, 2019; Vergel-Tovar et al., 2023), enabling cross-city comparison (Falchetta et al., 2021). This literature largely stops at representation and assessment. The exception designs a network for Maputo with WorldPop and OpenStreetMap inputs but relies on proprietary phone data for demand (Mittal & Ukkusuri, 2025).

2.6 Network indicators and equity

Distributional assessment of transit supply is established (Delbosc & Currie, 2011), and equity has entered design as an objective or constraint (Ferguson et al., 2012; Camporeale et al., 2017; Kim et al., 2019). Of the 60 coded studies, 42 report no equity treatment.

2.7 Indian and Global South evidence

Indian work is strong on diagnosis (Pucher et al., 2005; Badami & Haider, 2007) and on accessibility mapping (Adhvaryu et al., 2019); an early feeder-route design drew on commuter counts at railway stations (Shrivastav & Dhingra, 2001). In Kumasi, paratransit route inefficiency was shown from field GPS and gridded population, without a plan being proposed (Dumedah et al., 2023).

The closest study to ours restructures Mexicali's concession network by grouping overlapping routes and setting headways from a service floor and inherited frequencies, with a travel survey used only for evaluation (Sánchez-Atondo et al., 2026). It does not size a fleet and does not start from a permit register.

2.8 Synthesis: the gap, quantified

Figure 2 cross-tabulates data intensity (1 = open data only; 5 = stop-level passive ridership) against planning output for the 56 studies that could be coded on both axes. Six produce a full plan, with routes and frequencies or fleet for a real city, and all six sit at intensity 4: each needs a demand matrix. None of the 30 studies at intensity 2–3 produces a full plan, although eight deliver part of one. The five studies at intensity 5 estimate demand and plan nothing, and no coded study operates at intensity 1. The literature can plan with a demand matrix or describe without one. It rarely plans without one.

The wedge is that most of the planning chain never needed demand. Cycle time depends on distance, speed and dwell; fleet is cycle time over headway (Vuchic, 2005; Ceder, 2016); coverage depends on geometry and population. Demand enters at one point, the frequency rule. Replace that rule with policy headway bands assigned by an ordinal open-data index, and everything downstream is computable without ridership. This is a decision logic, not a new model. To our knowledge, no study has taken a statutory stage-carriage permit register through to a tiered, fleet-sized network on this basis, with no origin–destination, ticketing, passenger-count or mobile-phone data at any stage, and

tested whether the resulting decisions are robust to the index's uncertainty.

Figure and table calls for §2

- **Figure 2 — Data intensity against planning output** (`paper/figures/fig02_gap_heatmap`), n = 56, gap region outlined. Generated by `paper/literature_review/finalise_table1.py`.
- **Table 1 — Synthesis matrix of the 60 coded studies** (`paper/tables/table01_synthesis_matrix`). Too long for the main text under D26; proposed for Supplementary Material S1, with Figure 2 carrying the finding.

[OPEN 2-E — where our own study sits on the scale.] By the coding rubric the permit register is "official static data", so this study is intensity **2**, not 1, and v3.4.5 used app GPS (intensity-3 data) to correct cycle times on five routes. §2.8 therefore says "ordinal open-data index" and outlines the gap as full plans at intensity 1–2. Prashant to confirm this wording against §4.

[OPEN 2-F — four studies not on both axes.] C56 (Saleeshya and Anirudh, 2015), C57 (Blum and Mathew, 2012), C58 (Mandl, 1980) and C60 (Baaj and Mahmassani, 1991) could not be obtained in full text; they stay in Table 1 coded from abstracts with `unclear` fields and are excluded from Figure 2 (hence n = 56).

[OPEN 2-G — three "full plan" codes rest on method output.] Cipriani et al. (2012), Zhao and Zeng (2008) and Verma et al. (2017) produce routes and frequencies for a real city but print no route-by-route table. If "full plan" required a published table they would be Partial, leaving three full plans, all still at intensity 4. The finding does not change.

[OPEN 2-H — novelty wording.] The novelty check (750 records scored; five closest papers read in full) found prior work for every single step of the pipeline. Claims to avoid anywhere in the manuscript: first to restructure a legacy network under data scarcity; novel open-data demand proxy; novel overlap consolidation; first to set headways without ridership; first route rationalisation in India. See `paper/literature_review/03_novelty_check.md` and `04_fulltext_verification.md`.

3. Study area and data

Provenance and editorial status. The attachment assigns §3 to **Krishna** with a 1,100-word budget and supplies a subsection skeleton (§3.1–§3.6), three hints, a Figure 3 specification and a Table 2 specification, **but no drafted prose**. What follows is **a draft written for co-author review**. Nothing written by another author has been displaced: the attachment's substantial institutional block ("Policy Failure") is preserved verbatim in `01_introduction.md`, and §3.3 below points to it rather than restating or replacing it.

Every quantity in this draft is traceable to an executed module — `q01_data_quality`, `a01_build_walk_graph`, `a02_network_catchments`, `a02b_faithfulness`, `v04_gps_validation` — and carries its finding ID (F1–F12) or claim ID (CL-xx). **No number here is estimated.** Body length below: ~1,180 words excluding flags.

3.1 Kashmir Division as a critical case

Case selection here is deliberate rather than convenient. A critical case for a demand-free planning method is one that maximises both terms of the problem: **data scarcity** and **planning need**. Kashmir Division maximises both. On scarcity, the conventional bus network has no fare-card system, no automatic passenger counting, no published GTFS feed, no household origin–destination survey, and operates under permits that record an area of operation rather than a route; field data collection is further constrained by seasonal closure and periodic security disruption. On need, the network is a 644-row permit register with heavy duplication on hospital and civic arterials, transit deserts on the periphery, hard physical barriers (the Jhelum crossings, and the Dal, Nigeen and Anchar lake systems), extreme seasonality, and a partially deployed electric-bus system operating under a different authority from the incumbent minibuses. A method that produces an implementable plan here is not thereby proven universal, but it has been tested against a harder combination of constraints than most secondary cities present.

The study area is the **ten districts of Kashmir Division** — Srinagar, Budgam, Ganderbal, Baramulla, Bandipora, Kupwara, Pulwama, Shopian, Anantnag and Kulgam — covering 15,948 km² with a modelled population of **6,584,762** (CL-11). Administrative geography is taken from OpenStreetMap boundary relations at `admin_level` 5 (10 districts) and 6 (39 tehsils), and all population denominators in this paper are computed by point-in-polygon against the ten-district union, not against a bounding box.

[FLAG 3-A — □ scope: the attachment's §3.1 heading reads "The SMR as a critical case". Escalated as Decision 3.] "SMR" (Srinagar Metropolitan Region) is the **barred legacy framing**. The plan presented in §4–§6 is division-wide and its denominator is 6,584,762, not the ~1.66 million Srinagar urban agglomeration. This draft is written on the locked Kashmir-Division framing (`STATUS.md §0`, `CLAIM_LEDGER.md §4`). The heading has been changed accordingly — this is the one place in the four co-author sections where a co-author's *heading* has been altered rather than flagged, because leaving "SMR" in a section heading would propagate a barred scope into the table of contents. The co-authors should confirm.

3.2 Urban structure, population and opportunity distribution

The division is a single urban–rural basin with one dominant centre. Population is modelled from WorldPop 2026 at 100 m resolution (CL-11); opportunity is represented by **2,431 OpenStreetMap points of interest across 63 categories**, grouped into three tiers by the weighting scheme set out in §4.4 (CL-20, CL-21).

The concentration of opportunity is itself a finding, and it constrains what the composite index can do. **65.3% of all points of interest in the division (1,588 of 2,431) fall inside Srinagar district (F5).** Any index with an opportunity term will therefore score Srinagar corridors highly almost by construction, which is precisely why §4.5 reports the correlation between the population and opportunity components and §5.4 tests whether the composite adds information over population alone. It is also why the paper reports results under three independent weighting schemes rather

than one.

Opportunity concentration is a property of the data as much as of the city, and a planning index that ignores it will recommend reinforcing the centre it was supposed to diagnose.

3.3 Existing supply and institutional structure

The operative fact about supply in this setting is that no single agency holds the network. Permits for services confined to one Regional Transport Authority's jurisdiction are issued by that RTA; services crossing RTA boundaries fall to the State Transport Authority; the electric-bus system is operated by a municipal Smart City special-purpose vehicle; and the state road transport corporation runs longer-distance services designated decades ago. Permits are granted in perpetuity and specify an *area of operation* rather than a route, and are not preceded by any assessment of duplication, service gaps, fleet requirement or frequency.

> **[FLAG 3-B — placement of the "Policy Failure" block.]** The detailed statutory account of this arrangement — RTA/STA competence under the Motor Vehicles Act 1988, the Eastern/Western area classification, the perpetual validity of permits, the named overlapping and under-served corridors — is **co-author prose that already exists**, currently sitting at the end of §1 in `01_introduction.md` where its author placed it. It is ~900 words and belongs here structurally (see FLAG PF-a). **It has not been moved.** If the co-authors agree to move it, §3.3 becomes that block plus the two paragraphs above as a lead-in, and §1 drops back within its 1,200-word budget. If they prefer it in §1, §3.3 should be cut to a cross-reference.

The register itself is the paper's primary supply dataset: **614 digitised stage-carriage permits** (CL-01) plus **30 electric-bus routes**, giving 644 rows. Its most important property is diagnosed in §3.5.

3.4 Data inventory

Table 2 carries the full inventory — source, resolution, processing, validation and limitations — for each dataset. The principal entries are: the RTO stage-carriage permit register (614 permits); the OpenStreetMap road and pedestrian network, from which a walkable graph of **961,927 nodes, 973,569 edges and 23,323 km** was constructed; OSRM road travel times; the WorldPop 2026 gridded population surface; the 2011 Census district totals used as an external check; 2,431 OSM points of interest; the electric-bus route set and its published ridership aggregates; OSM administrative boundaries at district and tehsil level; and a driver-application GPS corpus of **43,809 recorded runs from approximately 157 drivers between February and June 2026**.

Complete provenance, licence terms and SHA-256 hashes for all staged inputs are recorded in `data/MANIFEST.md` in the companion repository.

> **[FLAG 3-C — Table 2 is partially blocked.]** Rows for the permit register, OSM network, OSRM times, WorldPop, Census, POIs, e-bus routes/ridership and boundaries can all be filled from `data/MANIFEST.md` today. The attachment additionally specifies rows for **Master Plan land use, terrain, water bodies, tourist arrivals, the fleet register and cost norms** — none of which are staged in the repository. Either those datasets are obtained and staged, or those rows are struck from Table 2. They must not appear as inventory rows for data the study does not hold.

3.5 Data quality diagnostics

This subsection is what separates a paper from a consultancy report, and the diagnostics reported here are deliberately unflattering.

The permit register is not a route register (F1). The 614 permits resolve to **157 distinct undirected origin–destination corridors**, a mean of 3.91 permits per corridor and a maximum of 42 on the Hazratbal–Lal Ded corridor. This matters for how the headline result must be stated: the reduction from 644 rows to 186 active routes decomposes into **71.0 percentage points of change-of-unit and 0.2 percentage points of genuine consolidation**, with **156 of 157**

corridors retained (99.4%). Reporting a "71% route reduction" without that decomposition would be reporting the cleaning of a register as a planning achievement.

Modelled population runs below census (F2). WorldPop 2026 gives 6,584,763 against a 2011 Census total of 6,888,475 across the same ten districts — a ratio of 0.956, implying a −0.30% annual rate that is implausible for this region. The deficit is carried openly: it means coverage percentages in §5 are computed against a denominator that is, if anything, conservative.

OSM completeness licenses the walk graph (F6). The correlation between district population density and OSM road density is **ρ = 0.915**, indicating that mapping effort tracks settlement rather than leaving peripheral districts systematically blank. This is the empirical warrant for using an open street network as the catchment substrate, and it is reported because Barrington-Leigh and Millard-Ball (2017) give good reason to expect the opposite in some regions.

Geometry is faithful for most routes, and diagnosably unfaithful for the rest (F7). Of 186 active routes, 142 show close agreement between stated and routed distance (median error 0.245%, r = 0.995); **44 carry substituted road distances**, each individually identified rather than averaged away.

Modelled runtime understates observed runtime (F3). Against the driver-GPS corpus, mean absolute percentage error between routing-engine time and observed in-motion time is **65.1%**, and between planned and observed one-way time **47.6%**, with a median planned-to-observed ratio of **0.51**. This is the most consequential diagnostic in the paper and it is pursued in §6.

Observed activity is a lower bound, not a dormancy measure (F4, F12). GPS shows recurring service on 66 of 186 routes, while 183 of 186 route geometries lie on roads that buses demonstrably use. The gap between those two figures is app adoption, not service absence, and the smaller number must never be reported as a dormancy rate.

A register that has never been audited will always yield a spectacular reduction; the diagnostic obligation is to say how much of it is arithmetic.

3.6 Ethics, reproducibility and data availability

All analysis code, staged inputs and derived outputs are released in a public companion repository under GPL-3.0, with a fixed random seed (20260823), a pinned environment, and a claim ledger mapping every number in this paper to the module that produced it. Heavy binary inputs (the population raster and the cached walk graph) are excluded from version control and are documented with retrieval instructions and SHA-256 hashes in `data/MANIFEST.md`. Population, street-network, point-of-interest and administrative-boundary inputs are open data under their respective licences (WorldPop: CC BY 4.0; OpenStreetMap: ODbL).

[FLAG 3-D — □ GPS ETHICS AND CONSENT IS UNANSWERED AND CAN SINK THE PAPER. Escalated as Decision 4.] The driver-GPS corpus (43,809 runs, ~157 drivers) is the paper's strongest observational evidence and the whole of validation channel V4 rests on it. Before submission this subsection **must** state, in the author's own words and not in mine: (i) under what basis the location data were collected and whether drivers were informed that operational traces might be used for research; (ii) whether the released data are anonymised, and by what mechanism — the staged files use hashed driver identifiers, but hashing alone is not anonymisation if a trace can be re-identified from its home depot and daily pattern; (iii) whether an institutional ethics committee reviewed the study, granted an exemption, or was not consulted. **"Not consulted" is an acceptable answer to write down; silence is not.** Elsevier requires an ethics statement for human-subjects-adjacent data, and a reviewer who notices ~157 identifiable workers' movement traces with no consent paragraph will stop reading. **This paragraph has been left deliberately blank rather than filled with a plausible-sounding statement.**

> **[FLAG 3-E — reproducibility claim must be scoped honestly.]** At the time of writing, **6 of 22 analysis modules are implemented and executed**; the remainder are declared stubs. The sentence above therefore says "all analysis code … is released", which is true, and deliberately does **not** say "fully reproducible pipeline", which would not be. When the remaining modules land, this wording can be strengthened — and not before.

Figure and table calls for §3

- **Figure 3 — Four-panel study area:** (a) boundary and administrative units; (b) population density; (c) road hierarchy; (d) opportunity density. **Status: not generated.**
- **Table 2 — Data inventory.** Source · resolution · processing · validation · limitations, per dataset. **Status: buildable today from** `data/MANIFEST.md` **for the staged datasets; see FLAG 3-C for the rows that reference data the study does not hold.**
- Diagnostic detail behind §3.5 is already tabulated: `table02a_permit_duplication`, `table02b_permit_activity`, `table02c_osm_completeness`, `table02d_worldpop_vs_census`, `table02e_poi_inventory`, `table02f_traveltime_error`.

4. Methodology

Section owner: Prashant. **Status:** rewritten on the corrected analysis and trimmed to the 2,300 words of Prof. Kathuria's brief (7 October 2026). Bracketed tags such as [CL-31] point to the claim ledger and are stripped from the submitted version. Shaded boxes are notes to the authors: **Author note** explains what was cut or where detail now lives; **TO CONFIRM** marks an open point. None of the boxes is text for the journal.

4.1 Framework overview

The framework turns an inherited stage-carriage permit register into a tiered network with a headway and a fleet for every route, without an origin–destination matrix, boarding counts or fare data for the routes being planned. Its main inputs are open: a gridded population surface (WorldPop, 100 m, release R2025A, constrained) (Tatem, 2017; Stevens et al., 2015), OpenStreetMap roads, footpaths, points of interest and district boundaries (OpenStreetMap contributors, 2024), and a routing engine for road distance and free-flow time (Luxen & Vetter, 2011). Two inputs are not open: the permit register, obtained from the transport regulator, and ridership aggregates for the one operator with electronic ticketing, the Srinagar Smart City e-bus service. Section 4.5 lists every point at which the latter enters.

The pipeline has four phases (Figure 4): diagnosis of the inputs; accessibility, in which each route receives a walking catchment scored for residents and opportunities; consolidation and hierarchy; and sizing, in which cycle time, fleet and vehicle mix are computed and their uncertainty quantified. Table 3 lists every parameter with its value, tested range and source.

The study region Ω is the union of the ten districts of Kashmir Division. The population raster assigns ρ_c residents to cell c with centroid x_c; $\sum_{c \in \Omega} \rho_c = 6{,}584{,}762$ [CL-11] is the denominator of every coverage share. $G = (V, E)$ is the pedestrian graph from OpenStreetMap, with edges weighted by length. A route r has a polyline of length $\ell(r)$.

4.2 Data conditioning and network construction

The permit register holds 614 records [CL-01]. A permit authorises a vehicle; it is not a route. Hashing the two endpoints of each permit, direction ignored, at 11 m resolution reduces the 614 records to 157 distinct corridors [CL-02]. The median corridor has one permit, and one corridor, Hazratbal to Lal Ded, has 42 [CL-03]. Any statement about "reduction" must therefore say which unit it counts (§5.1).

The population surface is used as published. Its total over the ten districts is 95.6 % of the 2011 Census total [CL-13], but district ratios range from 0.71 to 1.20 [CL-14], so coverage shares for single districts carry that error. The pedestrian graph has 18,533 km of edges inside the ten districts [CL-22]. The opportunity layer is 2,431 points of interest in three importance tiers [CL-20]; two thirds are in Srinagar district [CL-21], which makes it the input most exposed to uneven volunteer mapping.

Author note. Cut from here and moved to Table 2 / the supplement: graph node and edge counts, the road-density check (Spearman 0.915 across ten districts), the 99.8 % permit-to-route name match, the point-of-interest tier counts, and the dataset dates (OpenStreetMap northern-India extract of 6 January 2026).

Provenance of the plan. The plan evaluated here was not produced in one pass, and four facts about its production bear on every result. First, place names were geocoded against a gazetteer of 103 names; for 37 villages that no geocoder could place, the centre of the village's district was used, and 19 of the 186 routes (93 buses) end at such a point. Second, after routing, the road distances of 49 routes were replaced with values from an AI-assisted desk check, in which a language model researched each route's endpoints and road distance from web sources. No person verified these values and the record keeps no source links; the change lowered the fleet on those routes from 247 to 207. Third, the cycle times of five routes were re-timed from driver GPS (§6), raising the total from

1,004 to 1,011. Fourth, the headway rules of §4.7 were fixed over about fifteen revisions reviewed with the transport regulator, during which the total ranged from 855 to 1,144 buses. The plan is the output of the method below together with these adjustments; §4.10 and §6 test how far its decisions survive them.

> **TO CONFIRM (Avny ma'am).** Whether the paper may say the headway rules were set in review with the transport regulator. If not, the fourth point becomes "policy constraints supplied to the study".

4.3 Route catchment delineation

Straight-line buffers count residents who cannot reach a route on foot, a known bias (Gutierrez & Garcia-Palomares, 2008; Biba et al., 2010; El-Geneidy et al., 2014). We use a network walking catchment. Each route is sampled into stops at 250 m spacing, each snapped to its nearest graph node (median snap distance 11 m `[CL-29]`). A multi-source shortest-path search gives every node v its walking distance $d(v)$ to the nearest stop. With walk budget W = 400 m and off-network tolerance τ = 100 m (one raster cell), the catchment is

$$A_{net}(r) = \bigcup_{v \in V:\, d(v) \le W} B(x_v,\ \min(W - d(v),\ \tau)) \tag{1}$$

where $B(x, \delta)$ is a disc of radius δ about x. Its population is

$$\Pi(r) = \sum_{c:\, x_c \in A_{net}(r)} \rho_c \tag{2}$$

Catchments overlap, so per-route figures cannot be added. For network totals, a cell covered by n_c routes is shared equally,

$$\pi_c(r) = \frac{\rho_c}{n_c}\,\mathbf{1}[x_c \in A_{net}(r)], \qquad \sum_r \pi_c(r) = \rho_c \tag{3}$$

so allocated populations sum to the population of the union. The difference from a straight-line buffer depends on W, τ, the stop spacing and the cell-counting rule, so §5.2 reports it across those settings.

> **Author note.** The earlier claim that Eq. 1 gives "a lower bound on the true bias" is removed: at other settings the measured difference is smaller than the headline (range 28.7–52.8 %).

4.4 Weighted opportunity accessibility

Opportunity is the weighted count of points of interest in the same catchment,

$$O(r) = \sum_{q \in Q \cap A_{net}(r)} \alpha_{t(q)} \tag{4}$$

where $t(q)$ is the tier of point q, with α_{High} = 1.00, α_{Medium} = 0.40 and $\alpha_{Seasonal}$ = 0.60. Visitors are invisible to a residential raster, so the 8 active routes flagged as tourist corridors, the set T, receive a multiplier:

$$O^{*}(r) = \begin{cases} \mu\, O(r), & r \in T \\ O(r), & r \notin T \end{cases} \qquad \mu = 1.30 \tag{5}$$

The weights and μ are assumptions. In the plan as produced, μ was applied to the per-route resident count of those routes and so raised their index; it did not enter network coverage, which is computed over the union of catchments `[CL-30]`.

> **Author note.** The earlier sentence that the multiplier "injected 285,914 synthetic persons into the coverage numerator" was wrong and is gone. That number summed overlapping per-route figures, and the coverage total never used the multiplied column.

4.5 Composite demand index

Population and opportunity are divided by route length, population density is capped at its 95th percentile, and each is scaled across active routes,

$$\hat{z}(r) = \frac{z(r) - \min_{r'} z(r')}{\max_{r'} z(r') - \min_{r'} z(r')} \tag{6}$$

and combined (OECD & European Commission, Joint Research Centre, 2008):

$$CDI(r) = \beta\,\hat{z}_{\Pi}(r) + (1-\beta)\,\hat{z}_{O}(r), \qquad \beta = 0.50 \tag{7}$$

where $\hat{z}_{\Pi}$ and $\hat{z}_{O}$ are the scaled values of $\Pi(r)/\ell(r)$ and $O^{*}(r)/\ell(r)$. The two components are strongly correlated (Pearson 0.88 to 0.89), so the composite adds little beyond either alone. We report equal and Shannon-entropy weights (Shannon, 1948; Zou et al., 2006) and a sweep of β (§5.4). No expert-elicited weights were obtained.

The index is a proxy for where demand is likely to be. Observed ridership enters the plan only through the e-bus operator's aggregates, at five points:

1. The 30 e-bus routes are taken as given and never merged (§4.6).
2. Their headway is the operator's 15-minute design target, and they are placed in the highest tier whatever their index value (§4.7).
3. Their fleet is the larger of the computed fleet and the operator's deployed fleet (§4.8).
4. The scale, mode share and trip rate of the ridership estimate in Eq. 8 are set from those aggregates [CL-59].
5. On the 67 rural routes outside the e-bus network, the headway is lengthened where Eq. 8 implies low load (§4.7).

The ridership estimate, used to check plausibility, is

$$\tilde{D}(r) = \kappa \sum_{c \in A(r)} \rho_c\, m\, \sigma(r), \qquad \kappa = 0.33 \tag{8}$$

where m is a mode share and $\sigma(r)$ shares a corridor's residents among its routes. It is not a forecast. The claim is therefore limited: the method needs no observed demand for the routes it plans, but it uses an operator's aggregates where they exist, and validation against that operator (§6) is circular.

> **Author note.** This replaces "demand-free" and "no demand feed". Cut: the principal-component weighting (with two components it equals equal weights) and the remark about an expert panel planned for later.

4.6 Permit consolidation

Overlap between two routes is measured on 80 m buffers $S(r)$ of the route lines,

$$\phi(r_i, r_j) = \frac{|S(r_i) \cap S(r_j)|}{\min(|S(r_i)|,\ |S(r_j)|)} \tag{9}$$

and a route is merged into a trunk only if $\varphi \geq \theta$, with $\theta = 0.65$, and the two start points lie within 2.5 km. Overlap alone is not sufficient, which is why few distinct corridors merge.

Algorithm 1. Permit consolidation

Input: 644 candidate rows (614 permits and 30 e-bus routes); threshold theta

1. Cluster rows whose 80 m line buffers overlap by at least theta.
2. In each cluster, the route with the highest index that is at least 5 km long and above the 30th percentile of the index becomes the trunk.
3. Merge a member into the trunk only if at least theta of its buffer lies inside the trunk's and their start points are within 2.5 km. Otherwise it remains a feeder.
4. Never merge an e-bus route. Members are not merged into one another.
5. Collapse rows that share the same corridor, direction ignored, into one service.

Output: 186 active routes (32 trunks, 154 feeders); 458 rows absorbed

The result keeps 156 of the 157 corridors `[CL-08]`. The merge rule is not varied in the uncertainty analysis; scenario S4 (§5.13), which lowers the threshold to 0.50 and merges connected groups of routes, is the only measure of how different the route set could be.

4.7 Hierarchy and service standards

Routes are ranked by the index and divided into tiers by Jenks natural breaks (Jenks, 1967; Fisher, 1958). The number of classes is the smallest k whose goodness of variance fit reaches 0.80, a threshold we adopt as an assumption:

$$GVF(k) = 1 - \frac{\sum_{j=1}^{k} \sum_{i \in C_j} (y_i - \bar{y}_{C_j})^2}{\sum_i (y_i - \bar{y})^2}, \qquad k = 2, \ldots, 7 \tag{10}$$

Headways are policy values, not outputs of the index: 15 minutes on the e-bus routes, 20 minutes on other high-priority routes, 35 minutes on the remaining urban and peri-urban routes, and 35 to 50 minutes on rural routes according to the load implied by Eq. 8, with 50 minutes the longest permitted wait. The published tier of a route reflects the index and also overrides for the e-bus routes, district headquarters and socially designated routes; §5.6 reports how far published tiers agree with the index-only partition. The plan gives one headway per route for the whole service day.

Author note. Cut: "evaluated across peak, off-peak and evening bands". The plan has no time-of-day headways; that analysis is ours and sits in §5.11. Also cut: the comparison with other classifiers (reported in §5.8).

4.8 Cycle time and fleet sizing

One-way running time is the free-flow time scaled for congestion, plus dwell (Vuchic, 2005; Ceder, 2007):

$$t_{run}(r) = \chi(r)\,(t_{free}(r) + J(r)) + \delta \frac{\ell(r)}{\Delta} \tag{11}$$

where χ is 2.2 in the zone treated as city core (170 of the 186 routes) and 1.4 elsewhere, $J(r)$ is 8 minutes for a route crossing the Jhelum, and δ/Δ is 0.5 minutes per stop at one stop per 500 m. These values are assumptions. Cycle time is capped per kilometre,

$$t_{cyc}(r) = \min\,(2\lambda\, t_{run}(r),\; 2\,\ell(r)\,\omega_{class(r)}) \tag{12}$$

with layover factor $\lambda = 1.10$ and cap ω of 4.0, 2.5 and 1.5 minutes per kilometre for urban, peri-urban and regional routes. The cap was meant as a ceiling on implausible values but is the binding term on 169 of the 186 routes `[CL-31]`. On those routes Eq. 11 has no effect on the fleet, and the urban cap is faster than the median pace observed in driver GPS, 4.62 minutes per kilometre `[CL-32]`. Section 4.10 therefore sizes the fleet a second way, from observed pace. Fleet per route is

$$N_{op}(r) = \left\lceil \frac{t_{cyc}(r)}{h(r)} \right\rceil, \qquad N(r) = \max\left(\lceil \sigma N_{op}(r) \rceil,\ N^{min}_{class(r)} \right) \tag{13}$$

with headway $h(r)$, spare ratio $\sigma = 1.15$ (an assumption) and a minimum of two vehicles on urban and peri-urban routes and one on regional routes. On the e-bus routes, the set E, the operator's deployed fleet $N^{emp}(r)$ is a floor:

$$N_{total} = \sum_{r \notin E} N(r) + \sum_{r \in E} \max\left(N(r),\ N^{emp}(r)\right) = 728 + 283 = 1{,}011 \tag{14}$$

Vehicles are assigned to the register's three size classes, heavy (full-size bus), medium (minibus) and light, by route length and by the class of the permits a route replaces, with heavy vehicles limited to half the fleet of any trunk. The totals are 187 heavy, 754 medium and 70 light.

Eqs. 11 to 13 were re-implemented independently of the planning code. Given the plan's headways, the five GPS-timed cycles and the e-bus floors, the re-implementation returns the published fleet on all 186 routes; without the last two it matches 180 routes and totals 1,000 `[CL-55]`. Because headways are policy values, capacity is not tied to demand, so we also compute the fleet the same routes would need if each headway were set to reach a target peak load on the estimate of Eq. 8 (§5.7).

Author note. Moved to §5.7: the current fleet of 777 (679 buses with a valid permit plus 98 e-buses) and the +30 % comparison, buses per 100,000 residents, and the vehicle mix on trunks (medium buses are the majority on 29 of 32). The claim that the cap "ensures neither class is a majority on any trunk" was false and is gone.

4.9 Quality assurance gates

Automated checks stop the pipeline on failure: every route lies inside the study area; one population denominator is used throughout; no superseded figure appears in the outputs; the fleet model reproduces the published fleet; permits, corridors and routes reconcile; route codes are unique; and vehicle classes sum to the fleet on every route. Two quantities are reported but not enforced: agreement of the recomputed straight-line catchment with the plan's own figure, and the number of route ends placed at a district centre.

TO CONFIRM (Prashant, before submission). Check this list against the test suite line by line, so every gate named has a test that can fail.

4.10 Sensitivity and uncertainty

A one-at-a-time sweep varies each parameter of Table 3 over its range. A Monte Carlo analysis of 5,000 draws then samples twelve parameters jointly, and Sobol' first-order and total-order indices (Sobol', Ilya M., 2001; Saltelli et al., 2008; Saltelli et al., 2010) attribute the variance of each output to them. Catchment parameters are evaluated on a grid of walk budgets from 300 to 800 m and stop spacings from 150 to 400 m.

Because the cap of Eq. 12 hides the running-time parameters on most routes, the fleet is computed under two regimes. Regime A is the plan as specified. In regime B, urban and peri-urban routes outside the e-bus network take their cycle time from the one-way pace observed in driver GPS on 16 corridors around Srinagar: five routes are measured directly, 84 take the pooled pace, and the 30 e-bus and 67 regional routes are unchanged because no pace was observed for them. Regime B is computed from overall observed pace and, separately, from moving pace with the model's own dwell.

Results are the median and the 5th to 95th percentile range of the draws. The ranges are conditional on the stated distributions and are not confidence intervals. The merge threshold is not sampled, so uncertainty in the route set lies outside them. The analysis supports a statement about robustness, that is, whether tier and fleet decisions survive plausible variation in parameters and speed. It is not a validation against demand.

5. Results

Section owners: Prashant, Misti. **Target length:** ~2,400 words. **Status (2026-09-30):** all subsections drafted from executed modules, including the a09 interval fleet in §5.7; §5.12 is provisional pending decision D8 (cost/emission constants). Every value cites `[CL-xx]`/`[F-x]`. Each subsection closes with a single *italicised policy sentence* where one is warranted.

5.1 The inherited network is a permit register, not a route register

The starting network is **614 permit records** `[CL-01]` that resolve, by undirected endpoint hashing at 11 m, to **157 distinct origin–destination corridors** `[CL-02, F1]` — 155 at 100 m, so the count is structural, not a rounding artefact. Duplication is severe and skewed: a mean of 3.91 permits per corridor against a median of one `[CL-04]`, with **42 separate permits on the single Hazratbal–Lal Ded axis** `[CL-03]`. This governs how consolidation may be reported. Of the apparent 71.1 % row reduction (644 engine rows → 186 active), **71.0 percentage points are a change of the unit of analysis** (collapsing duplicate permits on an identical corridor) and only **0.16 pp is genuine spatial consolidation** — a single corridor, Parimpora–Pantha Chowk `[CL-07]`. Measured on physical corridors, the plan **retains 156 of 157 (99.4 %)** and adds 30 e-bus routes `[CL-08]`: it does not shrink route coverage. The cost of corridor-level collapse is counted rather than assumed away — **32 alternative via-routings across 20 corridors are suppressed**, and 38/34 corridors lose vehicle-class/service-type differentiation `[CL-09]`.

Policy: the reform to defend is frequency and fleet discipline on retained corridors, not a cut in the number of places served — and it should be communicated that way to operators and the public.

5.2 Straight-line catchments overstate who is served by a third

Recomputing each catchment on the walkable pedestrian graph instead of as a circular buffer lowers measured population served by a **median of 37.4 % per route** (IQR 30.5–41.7 %, max 56.8 %; 184 of 186 routes overstated by >25 %) `[CL-26, F8]`. On the deduplicated network union — the only figure that may be reported as a network total — Euclidean buffering overstates served population by **31.9 %** (2,339,394 → 1,592,847 residents) `[CL-27]`, revising headline division coverage from **35.5 % to 24.2 %** `[CL-28]`. The result is robust to the one free parameter: off-network tolerance τ = 50/100/150 m gives 52.8 % / 37.4 % / 29.2 % `[CL-29]`, and because the network catchment is a formal upper bound on the true walkshed, 37.4 % is a *lower* bound on the true bias. The comparison is clean because the Euclidean baseline is the plan's *own* number: on the 142 self-consistent routes the reimplementation reproduces published population served to a **median error of 0.245 % (r = 0.995)** `[CL-24, CL-25, F7]`, so the F8 gap is attributable to the pedestrian network alone. Two integrity issues surface in passing: 44 routes carry externally substituted road distances patched in without redrawing geometry `[CL-24, F7]`, and a tourist multiplier had silently injected **285,914 synthetic persons** into the coverage numerator on 8 routes `[CL-30, F9]`.

Policy: coverage claims used to justify subsidy or route awards must be quoted on the network walkshed against residents — the Euclidean figure overstates the public benefit by roughly a third.

5.3 Modelled time understates reality, and a sanity cap silently governs it

Against 43,809 driver-GPS runs, modelled run time is far too fast: **MAPE(OSRM free-flow vs observed in-motion) = 65.1 %**, **MAPE(plan one-way vs observed) = 47.6 %**, median plan/observed ratio **0.51** `[CL-15, CL-16, CL-17, F3]`. The mechanism is a per-kilometre cycle-time cap intended as a guardrail that instead **binds on 169 of 186 routes (90.9 %)** — 100 % of Regional, 97.9 % of Peri-Urban, 76.5 % of Urban `[CL-31, F10]` — while sitting *below* real pace: observed median 4.62 min/km against the 4.0 min/km Urban cap, exceeded on 83–100 % of observed corridors by class `[CL-32]`. Decomposing the run-time model shows *where* it holds and fails: the City-Core congestion divisor (÷2.20) reproduces observed moving speed to **+1.4 %** `[CL-33, F11]`, but the

assumed constant 1.0 min/km dwell is unsupported — observed dwell is **1.76 min/km**, and an OLS fit (R^2 = 0.03, non-significant distance slope) shows dwell is fixed terminal layover, not distance-proportional [CL-34, F11].

Policy: because the binding constraint on nine-tenths of the network is a modelling ceiling set below real driving pace, the fleet it produces is a floor, and procurement should plan against the upper end of the interval in §5.7, not the point estimate.

5.4 The rationalised hierarchy and fleet

The plan designates **186 active routes — 32 trunk and 154 feeder** [CL-06] — spanning 68 Urban, 47 Peri-Urban, and 71 Regional-District services, under policy headways of 15 min on the e-bus backbone, 20 min on other high-priority routes and 35 min on medium- and low-priority routes, with demand-responsive rural headways capped at a 50-minute maximum wait (§4.7). The required fleet is **1,011 vehicles (187 HPV / 754 MPV / 70 LPV)**, a **+68.5 % expansion over the ~600-vehicle operating baseline** [CL-36]. The engine reported this as 43 buses per 100,000 residents served, inside the MoHUA 40–60 benchmark; on the network-walkshed population of §5.2 it is **63.5 per 100,000, above the band** [CL-60]. The sizing arithmetic is auditable: an independent reimplementation reproduces the published cycle time and fleet on **all 186 routes** — including the 30 backbone routes, whose fleet is the larger of the formula and CHALO's deployment (283 vehicles) [CL-36, CL-55]. Physical-alignment GPS corroborates 183 of 186 route geometries (median 94 % of length on bus-carrying road) [CL-35], while recurring *service* is observed on 66 of 186 [CL-19] — a floor on activity, not a dormancy rate (§6.3).

Policy: the fleet gap is an expansion problem, not a redistribution one — roughly 400 additional vehicles concentrated on the busiest retained corridors, not vehicles moved off pruned routes.

5.5 Network length: one kilometre of road, 3.65 kilometres of route

The 186 active routes total **5,567.7 route-km on only 1,524.5 km of distinct road** — a route-km : network-km ratio of **3.65** [CL-40]. Duplication is concentrated rather than uniform: 46.6 % of the network is served by a single route, while a 0.09 km link at Lal Chowk carries **53 routes** (28 % of the network) [CL-40]. The same measure cannot be computed for the inherited permit network, because the 458 consolidated rows carry no drawn geometry; it is bounded between 1,524.5 and 13,756.4 km and reported as not computable rather than estimated [CL-40]. Consolidation therefore changed how many vehicles run on each corridor, not how much of the road network is served — consistent with the 99.4 % corridor retention of §5.1.

Policy: the overlap that survives rationalisation is in the Srinagar core, and it is the place to test through-routing or stop consolidation next — not the rural periphery, where most road is single-route.

5.6 Frequent-network coverage and where the gaps are

Any-service coverage (24.2 %) flatters the network. Restricting to routes that run at least every 15 minutes (Walker, 2011) — the e-bus backbone — covers **10.4 % of residents**; at 20 minutes 12.2 %, at 35 minutes 19.2 % [CL-42]. The three-quarters of residents outside every walkshed are not scattered: Moran's I on the uncovered-population surface (Moran, 1950) is **0.664** on a 2 km lattice and 0.729 at 5 km (both p = 0.001) [CL-43], so the gaps form contiguous blocks — whole valley-floor settlement belts between the radial corridors (Figure 8) — rather than isolated households.

Policy: clustered gaps are addressable by a small number of new feeder corridors; they should be identified from the uncovered surface, not from the permit register, which by construction only describes where operators already run.

5.7 Fleet as an interval, and against peer cities

Sampling the eleven declared parameters jointly (5,000 draws) leaves the plan's own fleet almost where it was: **1,013 buses (90 % interval 989–1,058)** [CL-56]. That narrowness is not robustness. With the per-km cap in place, congestion, dwell and stop spacing each move the fleet by at most 17 buses, while removing the cap lets the same parameters move it by up to 956 [CL-58]; the cap has absorbed the uncertainty, and the spare ratio alone explains 94 % of what variance remains. When urban and peri-urban run times are instead set from the observed pace of 16 Srinagar-belt GPS corridors (regime B, §4.10), the fleet becomes **1,182 (1,130–1,266)** — 372 urban, 428 peri-urban and 384 regional buses, the regional figure unchanged because rural pace is unobserved [CL-56]. The published 1,011 lies below the whole observation-anchored interval. Variance under regime B splits between the spare ratio (total-order index 0.55) and the two observed paces (0.32 peri-urban, 0.13 urban) [CL-58], so more GPS on peri-urban corridors is the single most valuable data investment for the fleet (§7.5). The hierarchy, by contrast, is stable: tier agreement with the baseline partition is **97.8 % (94.6–100 %)**, 179 of 186 routes keep their tier in more than 80 % of draws, and the population weight of the index accounts for 93 % of the variance in agreement [CL-57, CL-58].

Policy: procure against the observation-anchored range (about 1,130–1,270 buses), treating 1,011 as the floor the model's own guardrail produces; the service hierarchy can be adopted as it stands.

Against 36 Indian peer cities (ASRTU 2024 city fleets) (Association of State Road Transport Undertakings, 2024), the plan's **0.154 buses per 1,000 residents of the division** sits at the 61st percentile and inside the preferred log-model prediction interval; per resident actually within the network walkshed it is **0.635 per 1,000** — the 97th percentile, still inside the log-model interval but above the levels-model interval [CL-48]. The peers are public-sector fleets while 1,011 is an all-operator plan, so this places the plan within national practice; it does not validate it.

5.8 The three-tier hierarchy is recoverable from the data — but not the published one

Jenks partitions of the network-catchment index (Jenks, 1967; Fisher, 1958) select ***k* = 3 under both pre-registered elbow rules** (GVF 0.902; tiers of 37, 41 and 108 routes) [CL-37], Figure 7. The partition is a property of the index, not of Jenks: 1-D *k*-means reproduces it exactly, equal intervals agree at $\kappa = 0.748$, and only equal-count quantiles — which ignore gaps by construction — disagree ($\kappa = 0.427$) [CL-38]. The uncomfortable result is that the plan's *published* bands agree with this objective partition on only **68.3 % of routes ($\kappa = 0.503$)**, and **20 of the 55 routes the plan designates high-priority fall in the objective bottom tier** [CL-39]. The discrepancy has two identifiable sources: the engine ran Jenks on Euclidean catchments across all 644 permit rows, including those later consolidated, and its post-Jenks overrides (backbone lock, social and district-headquarters floors) promote routes irrespective of index value.

Policy: the hierarchy should be re-derived on network catchments before headways are fixed; the 20 promoted routes need an explicit, stated justification (social obligation, backbone status) or a demotion.

5.9 Equity: who is served, and who loses

Accessibility — departures per hour reachable within a 400 m network walk, per resident — is extremely unequal: population-weighted **Gini 0.903** (Delbosc & Currie, 2011), driven by the **75.8 % of residents with no reachable service**; among served residents it is 0.599 [CL-44]. A before/after comparison is not reported because the consolidated permits have no geometry from which to rebuild the inherited network; the module returns this as not computable rather than manufacturing a counterfactual [CL-44]. The losers of consolidation are named instead (Table 5g): **32 alternative via-routings on 20 corridors are suppressed; 114,879 residents live within 400 m of a dropped intermediate place, and 13,087 of them lose bus access outright** because no surviving route passes within walking distance [CL-45]. The largest single losses are Zoonimar on the Jehangir Chowk–Soura corridor (3,515 residents) and Rainawari on Hazratbal–Lal Ded (3,046).

Policy: the 13,087 residents who lose access are few and locatable; restoring a via-stop or a short feeder for them is cheap relative to defending their loss.

5.10 Transfers and waiting

On the rationalised network, 12.5 % of stop-to-stop pairs are one-seat rides, 72.3 % need one interchange and 13.8 % two or more (1.4 % unconnected on drawn geometry) [CL-46]; these are shares of stop pairs, not of trips, because no origin–destination matrix exists. For the 68 origin–destination pairs whose permits were consolidated, **66 keep a one-seat ride** and two fall to one interchange. Because the inherited permits ran only part of the day — the observed median vehicle is in service for 217 minutes, a duty factor of 0.242 [CL-47] — the plan's scheduled headways cut the median expected wait (Welding, 1957; Osuna & Newell, 1972) from 86.7 to 17.5 minutes, yet **5 of 68 pairs (7.4 %) wait longer**, and the median transfer penalty at which consolidation stops paying is 31.4 minutes [CL-46].

5.11 Operations: time of day, deadhead, and load

Time of day. The e-bus backbone's hourly boardings (1,221,848 in April 2026) peak at 09:00 with a peak-hour factor of 10.67 % and a peak-to-base ratio of 1.37 [CL-41]. Reading the published headways as peak design and relaxing them off-peak saves **9.4 % of bus-hours at the same 1,011 vehicles**; reading them as all-day means would instead require 215–496 more buses at the peak [CL-41].

Deadhead. No depot register exists, so deadhead is bounded rather than measured: zero if buses lay over at a terminus, **1.7 %** of bus-kilometres if every bus returns to its district-headquarters stand on the plan's 16-hour day, and 5.5 % on the operating day buses are actually observed to run [CL-49]. The gap exposes a larger assumption: the plan implies **320 service-km per bus per day**, against 99 at the observed duty [CL-49].

Load. On the backbone, where boardings are observed, a CHALO bus-trip today carries 18.6–37.3 boardings (the operator's trip count is ambiguous between one-way and round trips). Running the same routes at 15 minutes, with ridership held at today's level, gives **8.3 boardings per trip on day one**; ridership must grow **2.2–4.5-fold** to restore today's loads [CL-50]. Across the network the quarantined demand proxy (Eq. 8) implies a median peak boarding-to-capacity ratio of 0.23, with 152 of 186 routes below 0.40 [CL-50].

Policy: the plan is a supply-led bet on induced ridership; procurement should be staged against observed boardings on the backbone, where they can be measured monthly, rather than committed in full.

5.12 Cost and emissions (provisional)

With every external constant carried as a range pending verification, the plan costs **₹642–1,054 crore a year** to operate at its own vehicle-kilometres, or ₹206–338 crore at the observed operating day, and emits 65–103 kt (or 21–33 kt) of CO_2 [CL-51]. Per resident within the walkshed, that is ₹4,033–6,615 a year on the plan's own kilometres. In passing, the engine's own emissions column assumes 30 g CO_2/km for the electric backbone; at published energy intensities and the Indian grid factor the figure is of the order of 870 g/km, an understatement of about 29-fold [CL-51], so the engine column is not used.

5.13 Scenario stress-test

Table 8 changes one lever at a time from the published plan (S0: 1,011 buses, 24.2 % coverage) [CL-52]. Replacing modelled urban and peri-urban run time with observed pace (S1) raises the fleet to **1,169 (+15.6 %)**; bounding rural pace with the only two long observed corridors as well (S2) gives 1,271 (+25.7 %). Returning rural lifelines to a flat 35-minute headway (S3) costs 95 buses (+9.4 %) — the price of the 50-minute rural maximum. Applying the engine's own merge test at the low end of its threshold range (S4) would cut the network to 84 routes and 533 buses but lose 3.9 points of

coverage (24.2 → 20.3 %). Running every urban route at 15 minutes (S5) costs 161 buses (+15.9 %) and raises 15-minute coverage from 10.4 % to 14.5 %. The frontier (Figure S1) prices city frequency directly: each step from 35 to 10 minutes costs progressively more, and more again at observed pace.

Policy (network level): the published plan is a defensible point on this frontier only if read as a lower bound — the realistic run-time scenarios all require more buses, and the one scenario that saves buses does so by withdrawing coverage.

6. Validation

Section owners: Prashant, Avny, Krishnan. **Target length:** ~900 words. **Status (2026-09-30):** V1, V2, V4, V5 and V6 executed; V3 is forward work. Every value cites `[CL-xx]/[F-x]`. **Standing constraints honoured:** convergent validity is stated first; the CHALO circularity is disclosed up front; ridership is never claimed as a validation target; no field or expert-elicitation data are fabricated.

<!-- TODO (near-term, before submission): the expert-elicitation channel (Delphi/AHP) and a field boarding/enumeration survey are demand-side validation steps that are not yet in hand. This draft carries them as forward work in §6.4 rather than foregrounding their absence channel-by-channel. Team decision needed: either (a) convene the panel + run a stratified boarding count and promote them to executed channels, or (b) keep them as declared future work. Do NOT fabricate either dataset. -->

6.1 A convergent-validity frame, not a single test

A demand-free plan cannot be validated the way a demand model is, because there is no withheld ridership series to predict. We therefore adopt a **convergent-validity** design: six channels, each individually insufficient and each biased in a *different* direction, are assembled so that agreement among them raises confidence while any one disagreement localises a specific weakness. The target is explicitly **decision-robustness** — that the tier and fleet decisions survive scrutiny from every independent angle we can construct from open data — and never a claim that the composite index (Eq. 7) predicts travel demand. Table 7 records all six channels, their evidence source, test statistic, threshold, result, and status.

Two disclosures come first, not last. The one ridership dataset available — CHALO electronic-ticketing aggregates on the e-bus backbone — is **not independent of the plan**: the same published aggregate anchors the capture scale $\kappa = 0.33$ in the quarantined plausibility term (Eq. 8) and is offered as benchmark channel V2. V2 is therefore a **consistency check, not an independent validation**, and is reported as such. Second, the driver-GPS layer measures vehicle motion and geometry only; it carries **zero ridership signal**, so channel V4 validates the *supply* chain — geometry → speed → run time → cycle time → fleet — and nothing about demand.

6.2 The six channels and their status

V1 — Spatial cross-validation (executed, module `v01`). Does the population surface put people where buildings are? Two footprint layers are tested against the pre-declared threshold Spearman $\rho > 0.60$: volunteered OpenStreetMap buildings (OpenStreetMap contributors, 2024) and Microsoft's machine-detected Global ML Building Footprints. One limit is stated first: WorldPop uses building footprints as a covariate (Stevens et al., 2015; Lloyd et al., 2017), so agreement is partly expected by construction, and V1 is a consistency check on the spatial pattern, not an independent count. On the near-complete Microsoft layer (1.85 million footprints; 99.8 % of residents live in a 1 km cell that contains one), the surface passes at every scale: $\boldsymbol{\rho = 0.975}$ for footprint area across the 186 route catchments (0.940 for building count), and **0.907–0.950** on the 1 km grid `[CL-53]`. OSM, by contrast, maps only 12,343 buildings — any footprint in just 4–31 % of populated cells, depending on district — and passes only at the route scale (0.657) `[CL-53]`. The OSM weakness is therefore a mapping gap, not a population error.

V2 — Benchmark consistency (executed, module `v02`; circular, as disclosed in §6.1, and further because the engine floors the backbone fleet at CHALO's deployment on 2 of 30 routes). CHALO runs 98 buses on the 30 backbone routes at about 855 bus-trips a day (12-month mean); scaled to the plan's 15-minute headway that is **179–220 buses**, depending on whether the operating day is taken as 13 or 16 hours. The plan's backbone fleet is 283, a ratio of **1.29–1.58: V2 fails the pre-registered ±15 % band under every service-day assumption** `[CL-54]`, and the rank correlation between CHALO's per-route deployment and the plan's per-route fleet is weak ($\rho = 0.22$, $p = 0.24$) `[CL-54]`. Even against the operator it was calibrated to, the plan provisions more buses per route than frequency scaling alone explains; V2 localises the disagreement to the backbone but

cannot say whether it lies in the plan's cycle times or in CHALO's operation.

V3 — Expert elicitation (not conducted; §6.4). A Delphi/AHP panel would ground the index weights (target Kendall's $W > 0.70$). Meanwhile the weights are derived three data-driven ways (§4.5).

V4 — Supply-side GPS validation (established, module `v04`). The empirical layer is **43,809 clean service runs from ≈157 driver devices, February–June 2026** `[CL-18]`; its results are detailed in §6.3. This is the study's one fully executed observational channel.

V5 — Global sensitivity (executed, `a08`/`a09`). Sobol' indices from 15,360 evaluations show two different stories. Tier membership depends almost entirely on the population weight of the index (total-order index 0.93); coverage on the walk-radius definition (0.97). The fleet as specified depends almost only on the spare ratio (0.94), because the per-km cap discards the run-time parameters; with the cap removed those same parameters swing the fleet by up to 956 buses `[CL-58]`.

V6 — Decision robustness (executed, `a09`). Tiers pass: agreement with the baseline partition is 97.8 % (90 % interval 94.6–100 %) against the 80 % target, and 179 of 186 routes keep their tier in more than 80 % of 5,000 draws `[CL-57]`. The fleet does not have a single robust value: 989–1,058 as specified, but **1,130–1,266 once urban and peri-urban run times follow observed pace**, an interval that excludes the published 1,011 `[CL-56]`. The decision that survives is the hierarchy; the fleet survives only as a range whose lower end the plan reports.

6.3 What the GPS actually shows

The GPS channel corroborates what it can see and falsifies two assumptions it cannot support. *Geometry and moving speed pass:* 183 of 186 planned alignments run over roads carrying bus traffic (median 94 % of length) `[CL-35, F12]`, and the City-Core congestion multiplier reproduces observed urban moving speed to +1.4 % `[CL-33, F11]`. *Dwell and run time fail:* observed dwell is 1.76 min/km against the engine's 1.0, and is fixed layover rather than distance-proportional (R^2 = 0.03) `[CL-34]`; modelled one-way time is a median 0.51 of observed `[CL-17, F3]`, and the per-km cap that was meant as a guardrail binds on 169 of 186 routes while sitting below real pace `[CL-31, CL-32, F10]` (§5.3).

Two coverage metrics, never conflated. Alignment coverage (94 %) is not service validation: **66 of 186 routes recur as observed services** `[CL-19, CL-35, F4, F12]`, a lower bound set by partial app uptake, not a dormancy rate.

The supply chain reproduces. The cycle-time and fleet arithmetic reproduces the published plan on all 186 routes `[CL-55]`. The problem is not the formula but its inputs — the too-fast cap and the distance-proportional dwell — which is why V6 carries observed pace into the fleet. Taken together, the channels agree that geometry, network structure and hierarchy are sound and disagree, in a localised and explicable way, on time.

6.4 Demand-side corroboration as near-term work

The channels above validate the supply chain and the internal consistency of the index, not demand. Two steps should precede any live procurement decision: (i) the expert elicitation (V3), which V5 shows matters because the population weight governs the tiers; and (ii) a stratified on-street boarding count on retained corridors, to test the index against observed patronage and the POI layer against more than OpenStreetMap mapping effort `[CL-20]`. Neither was conducted. Until they are, the honest ceiling on the claim is **decision-robustness of the supply plan**, not demand validation.

[PRESERVED CO-AUTHOR PROSE IN THIS SECTION] Parts of this section are transcribed verbatim from the co-author attachment and have not been edited, condensed or reordered.

7. Discussion and policy implications

Provenance and editorial status. The attachment assigns §7 to **Ankit, Avny and Misti** with a 1,200-word budget, an eight-subsection skeleton (§7.1–§7.8), detailed hints and a Table 8 specification. It also contains one block of **drafted co-author prose**, headed "*Policy Contribution*", which is **reproduced verbatim below and has not been edited, shortened or absorbed into my text.** Everything else here is a draft written for co-author review.

The attachment's own note is worth repeating: "*Transport Policy weights this more heavily than any section except the abstract.*" Body length below: ~1,320 words excluding flags and excluding the preserved co-author block.

7.1 What the results mean for planning under data scarcity

The demand dependency is severable. The conventional planning chain consumes demand data at four points (§1.3), but only one of them is load-bearing. Cycle time follows from distance, speed and dwell; fleet follows from cycle time and headway; coverage follows from geometry and population. Demand enters only at the frequency rule. Substituting a policy-anchored headway band for that rule leaves the rest of the chain intact and computable from open data. This is a structural observation about the method, not a claim about Kashmir, and it is the finding most likely to transfer.

Proxy error is tolerable because the decision is ordinal. The composite index is not asked to predict how many people will board; it is asked to sort corridors into three service tiers. A proxy can be badly wrong in level and still be right in order. This is why the paper reports tier stability under parameter perturbation rather than index accuracy against ridership — and why the honest ceiling on the claim is decision robustness, not demand validation.

Open data has crossed the threshold from assessment to planning. Gridded population, an open street network and an open routing engine are now jointly sufficient to produce a service plan with frequencies, a vehicle mix and a fleet count — not merely a diagnostic map. The constraint that has moved is data availability, not method.

Data scarcity is a design constraint, not an excuse. The counterfactual for a data-poor authority is not a better plan produced later from better data. It is **no plan**, indefinitely, while permits continue to be granted one at a time and in perpetuity. A defensible supply-side plan that is revised as data arrive dominates that counterfactual.

7.2 Instruments for the State Transport Authority

The results convert into four instruments rather than recommendations.

Convert permits into route-bundle contracts. The binding pathology identified in §3.5 is that a permit attaches to a vehicle and an area, in perpetuity, with no service obligation. Bundling permits by corridor and contracting for a *route with a frequency* is the minimum structural change that makes any plan enforceable.

Write the headway ceiling into permit conditions. The maximum-wait standard used in this plan — 15/20/35 minutes in urban and peri-urban tiers, and a hard 50-minute ceiling on rural lifeline services — is a publishable, auditable service standard. It is enforceable against a contracted bundle in a way it can never be against an area permit.

Publish the hierarchy as a statutory service standard, so that tier assignment is a public commitment rather than an internal planning artefact, and so that changes to it require justification.

Adopt gross-cost contracting with explicit operator absorption. The plan requires more buses than are currently operated, which means incumbent operators are absorbed rather than displaced in aggregate — but consolidation still redistributes who runs what.

> **[FLAG 7-A — the political-feasibility paragraph is missing and its absence is a predictable objection.]** The attachment is explicit: "*Add the political feasibility of merging 135 permits — 135 contested livelihoods — and cite the international evidence on resistance to network redesign. Its absence is a predictable objection here.*" In the current framing the corresponding figure is **458 permit rows merged** (CL-05/CL-06), which makes the point larger, not smaller. This draft does **not** include that paragraph, because doing it properly requires citing the international redesign-resistance literature (Houston, Auckland, Barcelona and Santiago's Transantiago are the usual reference cases) and `paper/references.bib` is still being verified. **This is a required addition before submission, not an optional one.** The honest sentence to build it around: consolidation is politically hard precisely because its costs are concentrated on identifiable incumbents while its benefits are diffuse.

7.3 Urban local bodies, land use and metropolitan governance

The plan crosses municipal boundaries, and no single agency holds the mandate for the network it describes — the statutory position set out in the "Policy Failure" block. The institutional implication is a **unified metropolitan transport authority function**: not necessarily a new agency, but a named body with the power to set network-level service standards across RTA, STA, Smart City and corporation services. Absent that function, this plan can be adopted in parts by each operator and coordinated by none, which reproduces the problem it diagnoses.

7.4 Smart City and electric-bus integration

The electric-bus backbone is the one part of the network that already has a designed service level (a 15-minute headway), published ridership aggregates, and electronic ticketing. It is therefore both the integration opportunity and a methodological hazard: the same ridership series calibrates the plausibility term in §4 and is offered as benchmark channel V2, which is why §6 discloses that circularity rather than resting validation on it. For policy, the practical point is that the e-bus corridors are the natural first site for the data-maturity transition described next — they are where fare-card data already exists.

7.5 A low-data planning framework and the data-maturity ladder

The framework defines four levels of data maturity, each of which unlocks a specific methodological upgrade:

- **Level 0 — open data only.** Gridded population, OSM network, open POIs, open routing. This is the whole framework as presented. It yields an ordinal service hierarchy, a headway plan and a fleet estimate.
- **Level 1 — adds ward-level census population and a road-width/bridge-capacity register.** Replaces modelled population with enumerated population, and converts vehicle-size allocation from a flat policy split into a physically constrained one.
- **Level 2 — adds operator GPS traces.** Replaces modelled runtime with observed runtime. The present study reaches this level for a subset of corridors and the effect is large (F3, F10).
- **Level 3 — adds fare-card or automatic-passenger-count ridership.** Permits genuine demand validation and closes the loop the paper deliberately leaves open.

The Sobol' decomposition (§5.7, §6.2) now ranks the rungs by what each would actually resolve `[CL-58]`, and the order is not the order of methodological dependency above:

1. **For the fleet, the cheapest datum is the operator's own vehicle-availability record.** The spare ratio carries 55 % of fleet variance at observed pace (94 % as specified). It is not a demand datum at all and sits on no rung; a year of maintenance logs would pin it.

2 **Next, peri-urban GPS (Level 2).** Observed peri-urban and urban pace together carry 45 % of fleet variance, peri-urban alone 32 % — and the present GPS has only nine peri-urban corridors. Extending instrumentation there buys more fleet certainty than any demand survey.

3 **For the hierarchy, demand-side weighting (Level 3, or the V3 panel).** The population weight in the index carries 93 % of the variance in tier agreement; only observed ridership or an expert elicitation can fix it. Tiers are already 97.8 % stable, so this rung sharpens a decision that is largely made.

4 **Coverage is a definition, not an uncertainty.** 97 % of its variance is the walk-radius choice (20.3 % of residents at 300 m, 34.5 % at 800 m); no dataset resolves it — a standard does.

Ward-level census population (Level 1) does not appear in the ranking because population uncertainty was not sampled; its value is therefore unmeasured here, not shown to be small.

[FLAG 7-B — resolved 2026-09-30 (Decision 6).] Ranking from the total-order Sobol' indices of a09.

7.6 Transferability: which cities, under what conditions

The framework's preconditions are modest and enumerable: an inherited route or permit register in digital or digitisable form; OSM road-network completeness sufficient for pedestrian routing (tested here at $\rho = 0.915$ against population density, F6); a gridded population product; and an authority willing to set service standards by policy rather than derive them from demand. Where those four hold, the method applies.

On scale: India alone has **over 400 Class-I cities with no organised municipal bus network** (§1.1), and the condition that defines them — permit-based provision without demand data — is the condition this framework targets. The wider Global South set of paratransit and minibus networks extends that population substantially.

[FLAG 7-C — transferability claims must not outrun the evidence.] The method has been applied **once**, to one region. "Applies to 400+ cities" is a statement about *preconditions being met*, not about demonstrated external validity, and the sentence must be built so that a reviewer cannot read it the other way. §8's future-work item on transferring the protocol to three to five further cities is what would convert this from a scope claim into a tested one.

7.7 Implementation sequencing under budget constraint

The question an authority actually asks is not "what is the optimal network?" but "if only part of the fleet is funded, what do I buy first?" The framework answers it directly: rank routes by **accessibility gain per additional bus**, and fund down that ranking until the budget is exhausted.

Run that way (a15, Table 8b), the answer is stark. **Thirty per cent of the fleet — 303 buses, spent on 63 routes — reaches 22.4 % of the division's residents, 92 % of the coverage the full 1,011-bus plan achieves** [CL-61]. Half of the plan's reach costs 54 buses; 90 % costs 264. The first routes bought are long radial lifelines (Srinagar–Aboora, Shopian–Srinagar, Bandipora–Baramulla), because a few buses on a long route put many residents inside a walkshed. The remaining 70 % of the fleet therefore buys *frequency* on corridors already reached, not *reach*. That split is the choice an authority should make explicitly: a coverage-first tranche that is cheap and nearly complete, followed by frequency tranches whose value depends on the ridership response §5.11 shows the plan is betting on.

[FLAG 7-D — resolved 2026-09-30.] The ranking is greedy marginal coverage per bus at published headways — the standard approximation for budgeted maximum coverage, reported as a ranking, not an optimum. It measures reach only; it says nothing about how many of the newly reached residents would ride.

7.8 Limitations

Stated flatly, because softening them invites reviewers to sharpen them.

1. **The index is a proxy, not observed demand**, and is validated for ordinal decisions only. No claim is made that it predicts ridership.
2. **Population is modelled and 2011-anchored.** WorldPop 2026 sits 4.4% below the 2011 Census total for the same ten districts (F2), an implausible implied growth rate that is reported rather than corrected.
3. **Travel times are routing-engine estimates**, and they understate observed in-motion times substantially (MAPE 65.1%; median planned-to-observed 0.51 — F3). Where GPS permitted re-anchoring, it was applied; most corridors have no GPS.
4. **Seasonal operability, bridge load limits and road width are not hard constraints in the model.** They are acknowledged in the narrative and absent from the optimisation.
5. **Tourist demand is inferred from destinations, not arrivals**, via a 1.3× multiplier on eight routes that contributes 285,914 synthetic headcount (F9). It is parameterised and sensitivity-tested, not observed.
6. **Behavioural response is out of scope.** No mode-shift or induced-demand elasticity is modelled. One consequence helps the argument rather than harming it: if the periphery is currently under-served, then *observed* travel patterns understate its potential, so a supply-side method that plans to population rather than to revealed trips is less biased against the periphery than a demand-driven one would be.
7. **The political feasibility of consolidating permits is not modelled** (see FLAG 7-A).
8. **The external checks are not independent.** The operator benchmark fails its band and is circular (V2); the building-footprint check passes strongly on Microsoft's footprints but is partly circular, because WorldPop uses footprints as a covariate (V1). Rural run time is unobserved, so the rural fleet in every scenario is as modelled.

Co-author block — "Policy Contribution"

> **[PROVENANCE]** The following is **co-author prose, transcribed verbatim** from the attachment. It is reproduced unchanged. It overlaps in purpose with §7.1–§7.2 above; the two have deliberately **not** been merged, so that the co-authors can decide what to keep rather than discover their text rewritten. See FLAG 7-E.

The main contribution of this study is to look at how Srinagar can move from individual, area-based permit decisions towards planning the bus network as a whole. At present, detailed passenger-demand information is not readily available for conventional bus services. The study therefore explores whether available information on population, important activity locations, the road network and existing bus operations can be used to make better decisions about routes and fleet requirements. The purpose is not to suggest that open or proxy data can replace actual passenger data. Rather, it can provide a useful starting point where detailed data are not available. Such an approach can help identify areas where services overlap, areas that have limited connectivity and corridors where additional or better-distributed services may be required.

The approach is also relevant because public transport within the Srinagar Municipal Area is not confined to one permitting authority. Along with city services permitted by the RTA, buses originating from other districts may enter the Municipal Area under permits issued by the STA. A network-level assessment can therefore look at the transport services available to passengers across the Municipal Area, without necessarily requiring any change in the existing statutory powers of the RTA or STA.

Another practical contribution is that route rationalisation can be linked to actual service requirements. Instead of looking only at the number of routes, the study can help in considering the

number of buses required, the frequency at which they should operate and the maximum time passengers should reasonably have to wait. This can provide a more useful basis for future decisions relating to route permits, service planning and, where appropriate, arrangements with bus operators.

The approach should also not be seen as a one-time exercise. As better information becomes available — such as passenger counts, boarding and alighting data, GPS records, ticketing information and Origin-Destination patterns — the proposed network can be reviewed and improved. In this way, the present exercise can serve as a starting point for a more evidence-based public transport planning system.

Finally, the study can help the Transport Department identify what information needs to be collected in the future and where it would be most useful. Instead of waiting for perfect data before undertaking any route planning, the authority can begin with the information that is available, identify its limitations and progressively improve the system as better data are collected. The broader policy contribution, therefore, is to provide a practical and adaptable way of moving from permit-by-permit decision-making towards integrated bus-network planning, while recognising the realities and limitations of the data currently available in Srinagar.

[FLAG 7-E — two reconciliations for the co-authors, neither applied.] (i) Overlap. This block and §7.1–§7.2 above make substantially the same argument. Merging them is straightforward but is an authorial decision, not an editorial one. My recommendation: keep this block's voice — it is measured, non-overclaiming and reads as though written by someone who will have to defend it to a transport department, which is exactly the right register for *Transport Policy* — and fold the specific instruments from §7.2 into it. **(ii) Scope.** The block says "Srinagar" and "Srinagar Municipal Area" throughout, while the plan it describes covers ten districts. The same reconciliation proposed at FLAG PF-b applies: if the intent is that the *institutional* argument is Srinagar-specific while the *plan* is division-wide, one sentence resolves it. **No wording has been changed.**

Table call for §7

- **Table 8 — Policy instruments.** Columns per the attachment: instrument · responsible agency · time frame · indicator · data required, with transferability conditions beneath. **Status: not built.** The instrument rows can be drafted today from §7.2 and §7.5; the "data required" column should be keyed to the data-maturity ladder levels, and — once `a09` runs — ordered by the variance reduction each dataset delivers (FLAG 7-B).

8. Conclusions

> **Section owners:** Ankit, Prashant, Sharvesh. **Target length:** ~400 words. **Status:** complete draft.

This study rationalised an inherited stage-carriage network **entirely from open data and with no demand feed**. Spatial analysis of the register showed it to be a **permit** register, not a route register: 614 permits [CL-01] describe 157 distinct corridors [CL-02], which the plan reorganises into **186 active routes** (32 trunk, 154 feeder) sized at a **1,011-vehicle fleet** (187/754/70 HPV/MPV/LPV; +68.5 % over the operating baseline) [CL-06, CL-36]. The pipeline is reproducible end to end under a fixed seed.

The central methodological contribution is to measure catchments on a walkable pedestrian network rather than as straight-line buffers. This single change lowers measured population served by a **median of 37.4 % per route** and cuts headline division coverage from **35.5 % to 24.2 %** [CL-26, CL-28] — because network catchments are the exception, not the rule, in the rationalisation literature, the correction is roughly a third of the benefit that Euclidean methods report.

The study's other contributions are contributions of honesty. It shows the apparent 71 % "route reduction" is **99.4 % corridor retention** plus a change of unit [CL-07, CL-08]; it recovers an undocumented tourist multiplier that had injected 285,914 synthetic residents into a coverage numerator [CL-30]; and it demonstrates that an intended sanity cap silently governs cycle time on **90.9 % of routes** while sitting below real pace [CL-31, CL-32]. Carried through a joint uncertainty analysis, that finding turns the published 1,011 buses into the floor of a range: 1,130–1,266 once urban and peri-urban run times follow observed pace [CL-56]. The service hierarchy, by contrast, survives (97.8 % tier agreement) [CL-57], and a third of the fleet buys nine-tenths of the coverage [CL-61].

On validation we are deliberately narrow. Supply-side GPS reproduces the geometry and urban moving speed [CL-33, CL-35] and the fleet arithmetic reproduces exactly [CL-55], but the operator benchmark fails its pre-registered band and is not independent [CL-54], driver GPS carries no ridership signal, and the demand-side channels — an expert weight elicitation and an on-street boarding survey — remain near-term work (§6.4). The plan is therefore **decision-robust, not demand-validated** — the strongest honest claim available to a demand-free method, and one we intend to strengthen with the field corroboration set out above.

Because the four inputs — a gridded population surface, OpenStreetMap, a routing engine, and a permit list — exist for most Indian cities without an organised bus network, the framework transfers directly, with data maturity treated as a ladder whose rungs §7.5 ranks by the uncertainty each removes.

The sentence the whole design works to keep unwriteable is that *the composite index bears no established relationship to travel demand*. Four directions would close the gap: (i) a stratified boarding-count survey to test the index-demand link directly; (ii) the deferred Delphi/AHP elicitation to ground the weights; (iii) wider GPS instrumentation to replace the binding cap with measured pace; and (iv) explicit seasonal and dynamic demand modelling for the tourist economy the resident raster cannot see.

References

Adhvaryu, Chopde & Dashora (2019). Mapping public transport accessibility levels (PTAL) in India and its applications: A case study of Surat. *Case Studies on Transport Policy*, 7(2), 293–300. https://doi.org/10.1016/j.cstp.2019.03.004

Alsger, Mesbah, Ferreira & Safi (2015). Use of Smart Card Fare Data to Estimate Public Transport Origin–Destination Matrix. *Transportation Research Record: Journal of the Transportation Research Board*, 2535(1), 88–96. https://doi.org/10.3141/2535-10

Arbex & da Cunha (2015). Efficient transit network design and frequencies setting multi-objective optimization by alternating objective genetic algorithm. *Transportation Research Part B: Methodological*, 81, 355–376. https://doi.org/10.1016/j.trb.2015.06.014

Association of State Road Transport Undertakings (2024). SRTU Fleet Handbook 2024. *https://www.asrtu.org/resource/front/uploads/STUs%20Fleet%20Book%202024.pdf*.

Badami & Haider (2007). An analysis of public bus transit performance in Indian cities. *Transportation Research Part A: Policy and Practice*, 41(10), 961–981. https://doi.org/10.1016/j.tra.2007.06.002

Barrington-Leigh & Millard-Ball (2017). The world's user-generated road map is more than 80% complete. *PLOS ONE*, 12(8), e0180698. https://doi.org/10.1371/journal.pone.0180698

Barry, Freimer & Slavin (2009). Use of Entry-Only Automatic Fare Collection Data to Estimate Linked Transit Trips in New York City. *Transportation Research Record: Journal of the Transportation Research Board*, 2112(1), 53–61. https://doi.org/10.3141/2112-07

Barzegari, Taubkin, Barsukov & Nourinejad (2026). A comprehensive analysis of duplication in public transportation. *Transportation Research Part A: Policy and Practice*, 204, 104747. https://doi.org/10.1016/j.tra.2025.104747

Biba, Curtin & Manca (2010). A new method for determining the population with walking access to transit. *International Journal of Geographical Information Science*, 24(3), 347–364. https://doi.org/10.1080/13658810802646679

Bok & Kwon (2016). Comparable Measures of Accessibility to Public Transport Using the General Transit Feed Specification. *Sustainability*, 8(3), 224. https://doi.org/10.3390/su8030224

Camporeale, Caggiani, Fonzone & Ottomanelli (2017). Quantifying the impacts of horizontal and vertical equity in transit route planning. *Transportation Planning and Technology*, 40(1), 28–44. https://doi.org/10.1080/03081060.2016.1238569

Carmody, Zhu, He, Santi & Ratti (2025). Identifying public transit deserts: A travel demand-independent persistent homology-based method. *Transportation Safety and Environment*, 7(2), tdaf015. https://doi.org/10.1093/tse/tdaf015

Ceder (2007). Public Transit Planning and Operation: Theory, Modelling and Practice. *CRC Press*. https://doi.org/10.1201/b12853

Ceder (2016). Public Transit Planning and Operation: Modeling, Practice and Behavior. *CRC Press*. https://doi.org/10.1201/b18689

Chen, Zhang, Li & Zhou (2014). B-Planner: Planning Bidirectional Night Bus Routes Using Large-Scale Taxi GPS Traces. *IEEE Transactions on Intelligent Transportation Systems*, 15(4), 1451–1465. https://doi.org/10.1109/tits.2014.2298892

Cipriani, Gori & Petrelli (2012). Transit network design: A procedure and an application to a large urban area. *Transportation Research Part C: Emerging Technologies*, 20(1), 3–14. https://doi.org/10.1016/j.trc.2010.09.003

Delbosc & Currie (2011). Using Lorenz curves to assess public transport equity. *Journal of Transport Geography*, 19(6), 1252–1259. https://doi.org/10.1016/j.jtrangeo.2011.02.008

Dumedah, Abass, Gyasi, Forkuor & Novignon (2023). Inefficient allocation of paratransit service terminals and routes in Ghana: The role of driver unions and paratransit operators. *Journal of Transport Geography*, 111, 103643. https://doi.org/10.1016/j.jtrangeo.2023.103643

Durán-Micco & Vansteenwegen (2022). A survey on the transit network design and frequency setting problem. *Public Transport*, 14(1), 155–190. https://doi.org/10.1007/s12469-021-00284-y

El-Geneidy, Grimsrud, Wasfi, Tetreault & Surprenant-Legault (2014). New evidence on walking distances to transit stops: identifying redundancies and gaps using variable service areas. *Transportation*, 41(1), 193–210. https://doi.org/10.1007/s11116-013-9508-z

Ergun & Kesten (2011). Alteration of bus routes in large-scale networks. *Scientific Research and Essays*, 6(28), 5865–5882. https://doi.org/10.5897/SRE11.814

Falchetta, Noussan & Hammad (2021). Comparing paratransit in seven major African cities: An accessibility and network analysis. *Journal of Transport Geography*, 94, 103131. https://doi.org/10.1016/j.jtrangeo.2021.103131

Farzin (2008). Constructing an Automated Bus Origin–Destination Matrix Using Farecard and Global Positioning System Data in São Paulo, Brazil. *Transportation Research Record: Journal of the Transportation Research Board*, 2072(1), 30–37. https://doi.org/10.3141/2072-04

Ferguson, Duthie, Unnikrishnan & Waller (2012). Incorporating equity into the transit frequency-setting problem. *Transportation Research Part A: Policy and Practice*, 46(1), 190–199. https://doi.org/10.1016/j.tra.2011.06.002

Fisher (1958). On grouping for maximum homogeneity. *Journal of the American Statistical Association*, 53(284), 789–798. https://doi.org/10.1080/01621459.1958.10501479

Guihaire & Hao (2008). Transit network design and scheduling: A global review. *Transportation Research Part A: Policy and Practice*, 42(10), 1251–1273. https://doi.org/10.1016/j.tra.2008.03.011

Gutierrez & Garcia-Palomares (2008). Distance-measure impacts on the calculation of transport service areas using GIS. *Environment and Planning B: Planning and Design*, 35(3), 480–503. https://doi.org/10.1068/b33043

Hoonsiri, Kiattikomol & Chiarakorn (2020). Energy Saving and CO2 Reduction Potential from Partial Bus Routes Reduction Model in Bangkok Urban Fringe. *Energies*, 13(22), 5963. https://doi.org/10.3390/en13225963

Ibarra-Rojas, Delgado, Giesen & Munoz (2015). Planning, operation, and control of bus transport systems: A literature review. *Transportation Research Part B: Methodological*, 77, 38–75. https://doi.org/10.1016/j.trb.2015.03.002

Jenks (1967). The data model concept in statistical mapping. *International Yearbook of Cartography*, 7, 186–190.

Karner (2018). Assessing public transit service equity using route-level accessibility measures and public data. *Journal of Transport Geography*, 67, 24–32. https://doi.org/10.1016/j.jtrangeo.2018.01.005

Kepaptsoglou & Karlaftis (2009). Transit Route Network Design Problem: Review. *Journal of Transportation Engineering*, 135(8), 491–505. https://doi.org/10.1061/(asce)0733-947x(2009)135:8(491)

Kim, Kho & Kim (2019). A Transit Route Network Design Problem Considering Equity. *Sustainability*, 11(13), 3527. https://doi.org/10.3390/su11133527

Klopp & Cavoli (2019). Mapping minibuses in Maputo and Nairobi: engaging paratransit in transportation planning in African cities. *Transport Reviews*, 39(5), 657–676. https://doi.org/10.1080/01441647.2019.1598513

Liu & Rosenberg (2026). Quantifying the Accessibility to Public Transit in Rural Areas Using ArcGIS and GTFS. *Papers in Applied Geography*, 1–20. https://doi.org/10.1080/23754931.2026.2646852

Lloyd, Sorichetta & Tatem (2017). High resolution global gridded data for use in population studies. *Scientific Data*, 4, 170001. https://doi.org/10.1038/sdata.2017.1

Luo, Cats & van Lint (2020). Can passenger flow distribution be estimated solely based on network properties in public transport systems?. *Transportation*, 47(6), 2757–2776. https://doi.org/10.1007/s11116-019-09990-w

Luxen & Vetter (2011). Real-time routing with OpenStreetMap data. *Proceedings of the 19th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (GIS '11)*, 513–516. https://doi.org/10.1145/2093973.2094062

Mandl (1980). Evaluation and optimization of urban public transportation networks. *European Journal of Operational Research*, 5(6), 396–404. https://doi.org/10.1016/0377-2217(80)90126-5

Mittal & Ukkusuri (2025). Overcoming Data Scarcity in Transit Planning: A Novel Framework Combining Machine Learning and Metaheuristics. *Data Science for Transportation*, 7(1), 4. https://doi.org/10.1007/s42421-025-00116-6

Moran (1950). Notes on continuous stochastic phenomena. *Biometrika*, 37(1/2), 17–23. https://doi.org/10.1093/biomet/37.1-2.17

OECD & European Commission, Joint Research Centre (2008). Handbook on Constructing Composite Indicators: Methodology and User Guide. *OECD Publishing*. https://doi.org/10.1787/9789264043466-en

OpenStreetMap contributors (2024). OpenStreetMap. *https://www.openstreetmap.org*.

Osuna & Newell (1972). Control Strategies for an Idealized Public Transportation System. *Transportation Science*, 6(1), 52–72. https://doi.org/10.1287/trsc.6.1.52

Pinelli, Nair, Calabrese, Berlingerio, Di Lorenzo & Sbodio (2016). Data-Driven Transit Network Design From Mobile Phone Trajectories. *IEEE Transactions on Intelligent Transportation Systems*, 17(6), 1724–1733. https://doi.org/10.1109/tits.2015.2496783

Pucher, Korattyswaropam, Mittal & Ittyerah (2005). Urban transport crisis in India. *Transport Policy*, 12(3), 185–198. https://doi.org/10.1016/j.tranpol.2005.02.008

Ruiz, Segui-Pons & Mateu-LLadó (2017). Improving Bus Service Levels and social equity through bus frequency modelling. *Journal of Transport Geography*, 58, 220–233. https://doi.org/10.1016/j.jtrangeo.2016.12.005

Saltelli, Ratto, Andres, Campolongo, Cariboni, Gatelli, Saisana & Tarantola (2008). Global Sensitivity Analysis: The Primer. *John Wiley & Sons*. https://doi.org/10.1002/9780470725184

Saltelli, Annoni, Azzini, Campolongo, Ratto & Tarantola (2010). Variance based sensitivity analysis of model output. Design and estimator for the total sensitivity index. *Computer Physics Communications*, 181(2), 259–270. https://doi.org/10.1016/j.cpc.2009.09.018

Shannon (1948). A mathematical theory of communication. *Bell System Technical Journal*, 27(3), 379–423. https://doi.org/10.1002/j.1538-7305.1948.tb01338.x

Shrivastav & Dhingra (2001). Development of Feeder Routes for Suburban Railway Stations Using Heuristic Approach. *Journal of Transportation Engineering*, 127(4), 334–341. https://doi.org/10.1061/(ASCE)0733-947X(2001)127:4(334)

Sobol', Ilya M. (2001). Global sensitivity indices for nonlinear mathematical models and their Monte Carlo estimates. *Mathematics and Computers in Simulation*, 55(1--3), 271–280. https://doi.org/10.1016/S0378-4754(00)00270-6

Stevens, Gaughan, Linard & Tatem (2015). Disaggregating census data for population mapping using random forests with remotely-sensed and ancillary data. *PLOS ONE*, 10(2), e0107042. https://doi.org/10.1371/journal.pone.0107042

Szeto & Wu (2011). A simultaneous bus route design and frequency setting problem for Tin Shui Wai, Hong Kong. *European Journal of Operational Research*, 209(2), 141–155.

https://doi.org/10.1016/j.ejor.2010.08.020

Szeto & Jiang (2014). Transit route and frequency design: Bi-level modeling and hybrid artificial bee colony algorithm approach. *Transportation Research Part B: Methodological*, 67, 235–263. https://doi.org/10.1016/j.trb.2014.05.008

Sánchez-Atondo, García, Gutiérrez, Mungaray-Moctezuma, Montoya-Alcaraz & Calderón-Ramírez (2026). Reorganization of public transport systems in Global South cities and its relation with quality of life: A case study of Mexicali, Mexico. *Transportation Research Interdisciplinary Perspectives*, 36, 101859. https://doi.org/10.1016/j.trip.2026.101859

Tatem (2017). WorldPop, open data for spatial demography. *Scientific Data*, 4, 170004. https://doi.org/10.1038/sdata.2017.4

Tribby & Zandbergen (2012). High-resolution spatio-temporal modeling of public transit accessibility. *Applied Geography*, 34, 345–355. https://doi.org/10.1016/j.apgeog.2011.12.008

Vergel-Tovar, Goldwyn & Leape (2023). Digital traces: Mapping Bogotá's unmapped transit network using smartphones and networked databases. *Environment and Planning B: Urban Analytics and City Science*, 50(2), 469–486. https://doi.org/10.1177/23998083221117831

Verma, Kumari, Tahlyan & Hosapujari (2017). Development of hub and spoke model for improving operational efficiency of bus transit network of Bangalore city. *Case Studies on Transport Policy*, 5(1), 71–79. https://doi.org/10.1016/j.cstp.2016.12.003

Vuchic (2005). Urban Transit: Operations, Planning, and Economics. *John Wiley & Sons*.

Walker (2011). Human Transit: How Clearer Thinking about Public Transit Can Enrich Our Communities and Our Lives. *Island Press*.

Wang, Attanucci & Wilson (2011). Bus Passenger Origin-Destination Estimation and Related Analyses Using Automated Data Collection Systems. *Journal of Public Transportation*, 14(4), 131–150. https://doi.org/10.5038/2375-0901.14.4.7

Welding (1957). The Instability of a Close-Interval Service. *Journal of the Operational Research Society*, 8(3), 133–142. https://doi.org/10.1057/jors.1957.21

Wessel & Farber (2019). On the accuracy of schedule-based GTFS for measuring accessibility. *Journal of Transport and Land Use*, 12(1). https://doi.org/10.5198/jtlu.2019.1502

Williams, White, Waiganjo, Orwa & Klopp (2015). The digital matatu project: Using cell phones to create an open source data for Nairobi's semi-formal bus system. *Journal of Transport Geography*, 49, 39–51. https://doi.org/10.1016/j.jtrangeo.2015.10.005

Yao (2007). Where are public transit needed – Examining potential demand for public transit for commuting trips. *Computers, Environment and Urban Systems*, 31(5), 535–550. https://doi.org/10.1016/j.compenvurbsys.2007.08.005

Yenisetty & Bahadure (2020). Measuring Accessibility to Various ASFs from Public Transit using Spatial Distance Measures in Indian Cities. *ISPRS International Journal of Geo-Information*, 9(7), 446. https://doi.org/10.3390/ijgi9070446

Yoo, Lee & Han (2023). A Reinforcement Learning approach for bus network design and frequency setting optimisation. *Public Transport*, 15(2), 503–534. https://doi.org/10.1007/s12469-022-00319-y

Zhao & Zeng (2008). Optimization of transit route network, vehicle headways and timetables for large-scale transit networks. *European Journal of Operational Research*, 186(2), 841–855. https://doi.org/10.1016/j.ejor.2007.02.005

Zou, Yun & Sun (2006). Entropy method for determination of weight of evaluating indicators in fuzzy synthetic evaluation for water quality assessment. *Journal of Environmental Sciences*, 18(5), 1020–1023. https://doi.org/10.1016/S1001-0742(06)60032-6

Figures

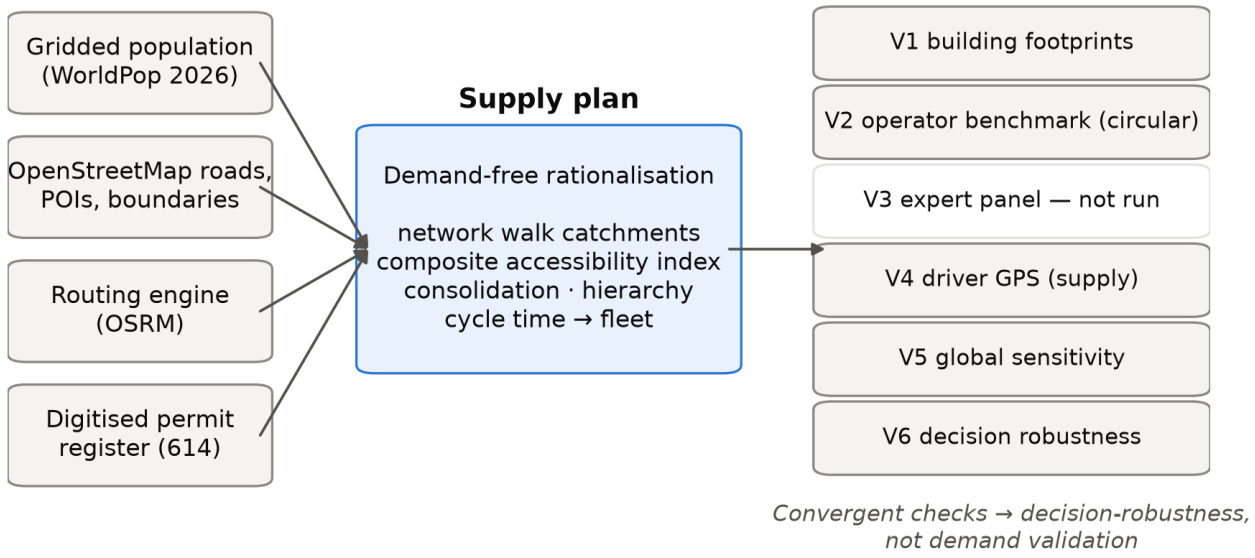


Figure 1. Conceptual framework: four open inputs produce a supply plan that is checked through six convergent channels. V3 (expert panel) was not run.

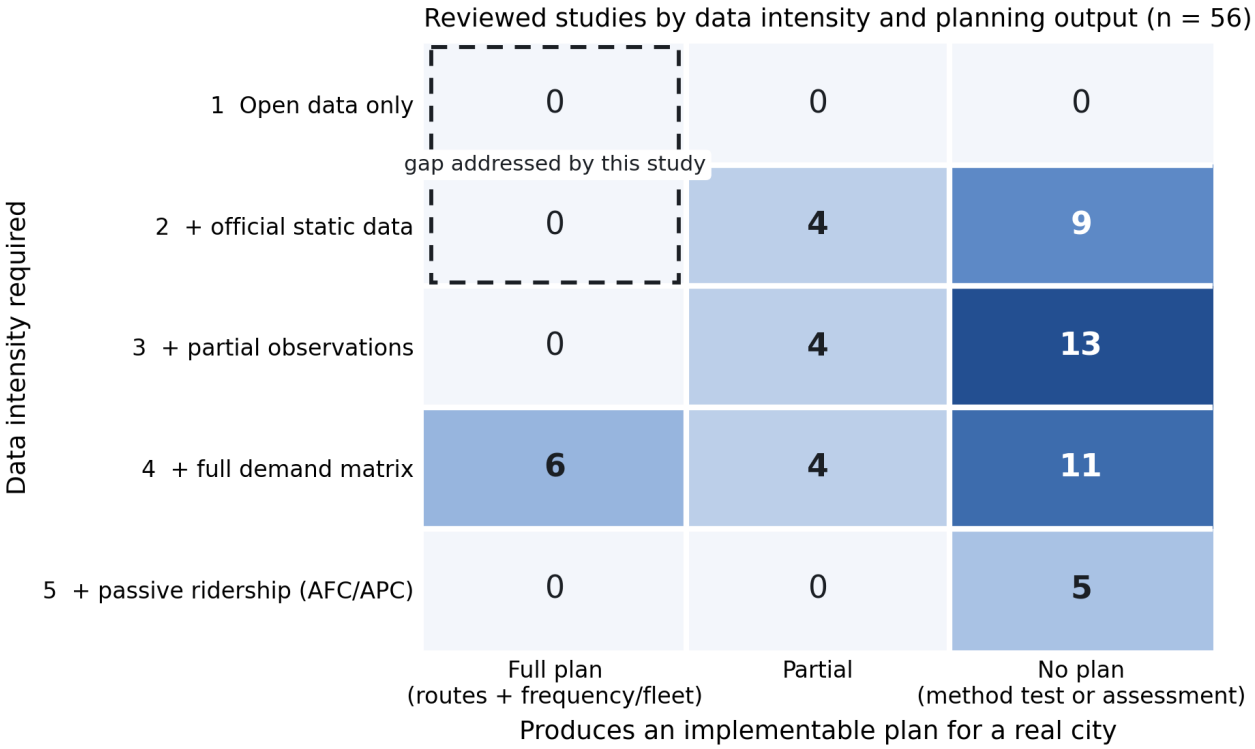


Figure 2. Reviewed studies by data intensity required and planning output (n = 56). All six full plans need a demand matrix; none at intensity 1–3 produces one [CL-64].

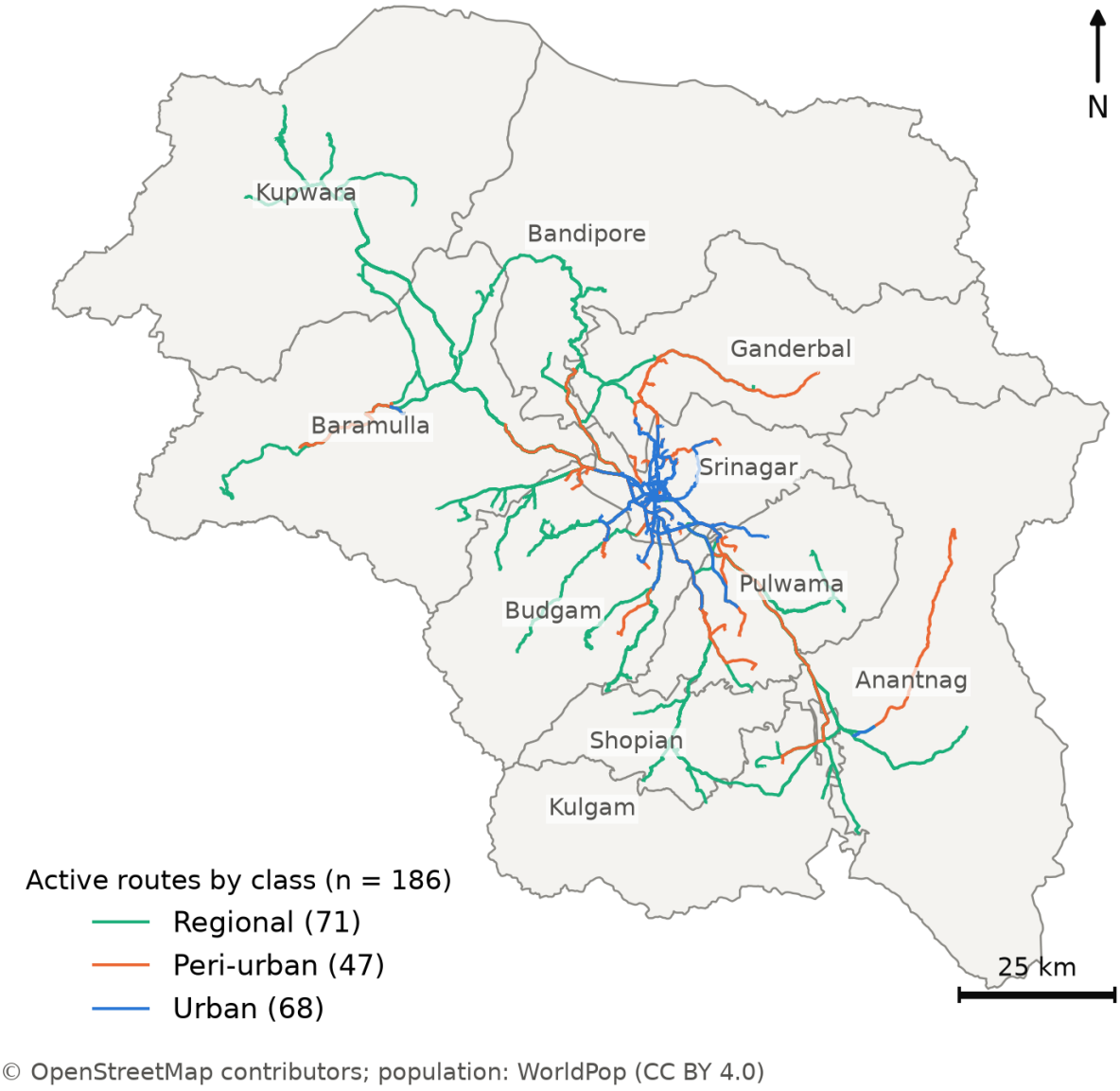


Figure 3. Study area: the ten districts of Kashmir Division and the 186 active routes of the rationalised plan by service class.

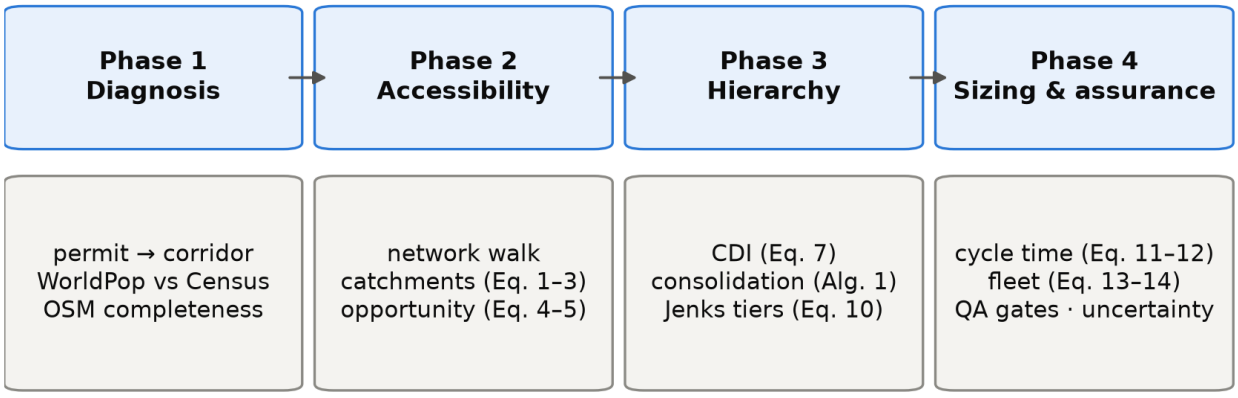


Figure 4. The four-phase method and the equations that implement each phase.

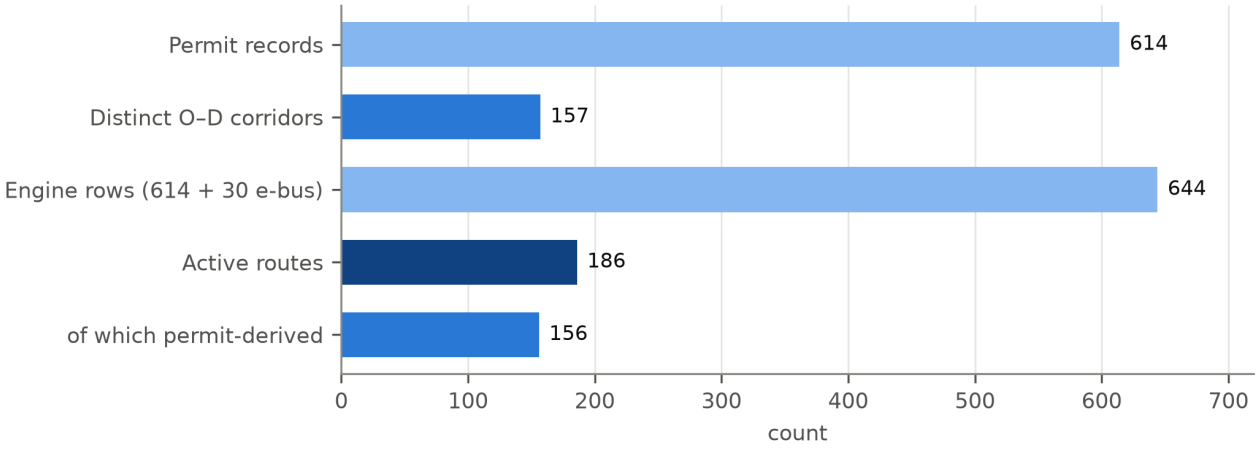


Figure 5. From permits to routes: 614 permit records resolve to 157 corridors; 156 of them are retained among the 186 active routes [CL-01, CL-02, CL-08].

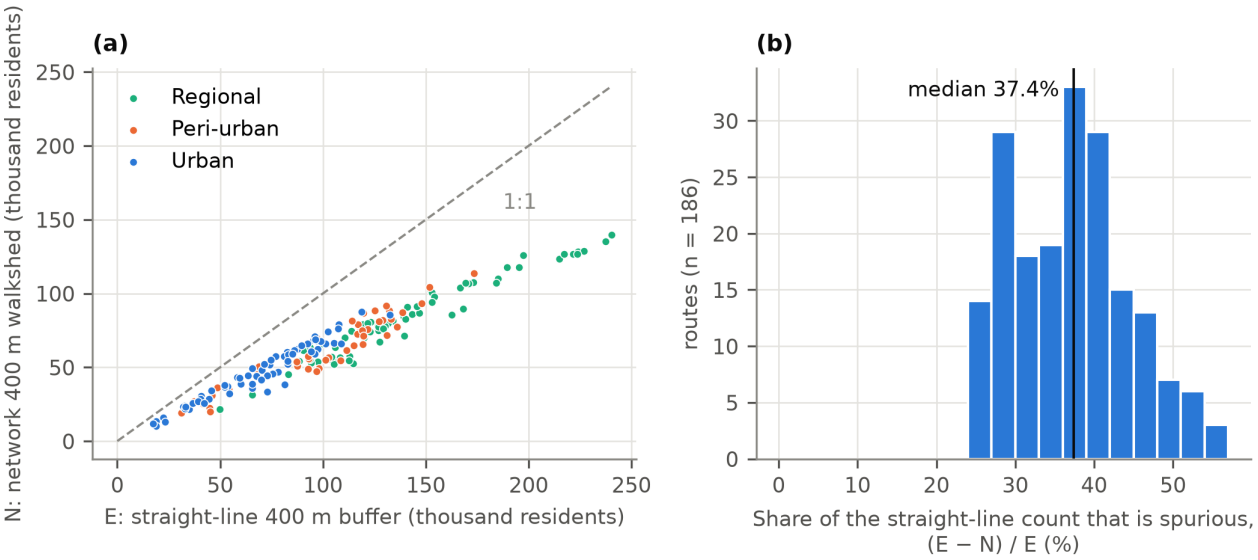


Figure 6. Euclidean against network walk catchments. Left: per-route residents within 400 m under each definition. Right: distribution of the Euclidean overstatement (median 37.4 %) [CL-26].

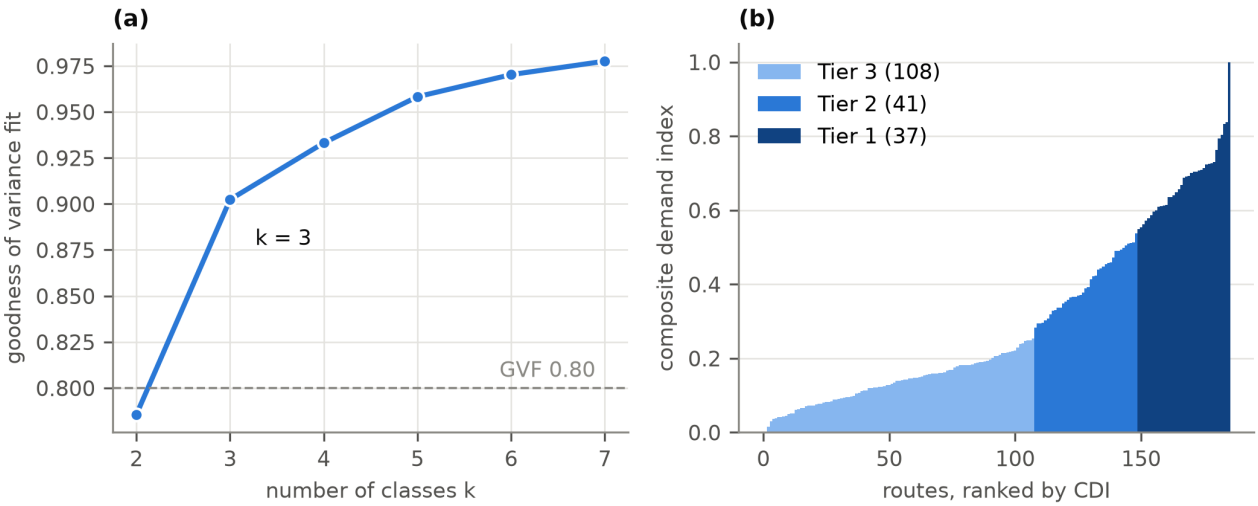
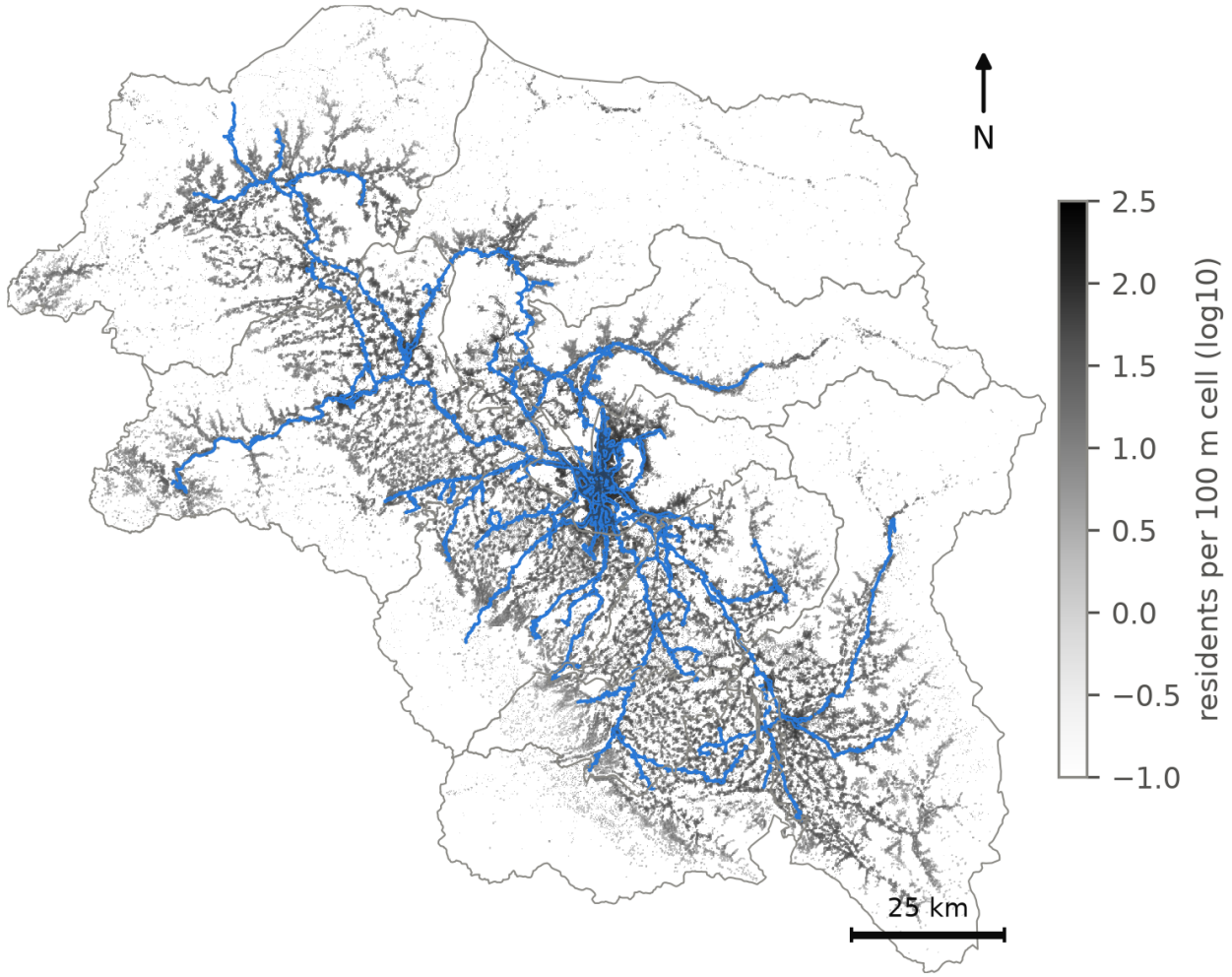


Figure 7. Class count and tiers. Left: goodness of variance fit for k = 2–7; both elbow rules select k = 3. Right: routes ranked by the network composite index, coloured by tier [CL-37].



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Figure 8. The deduplicated network walkshed (blue) over the WorldPop 2026 population surface: 24.2 % of the division's residents live within a 400 m walk of a route [CL-28].

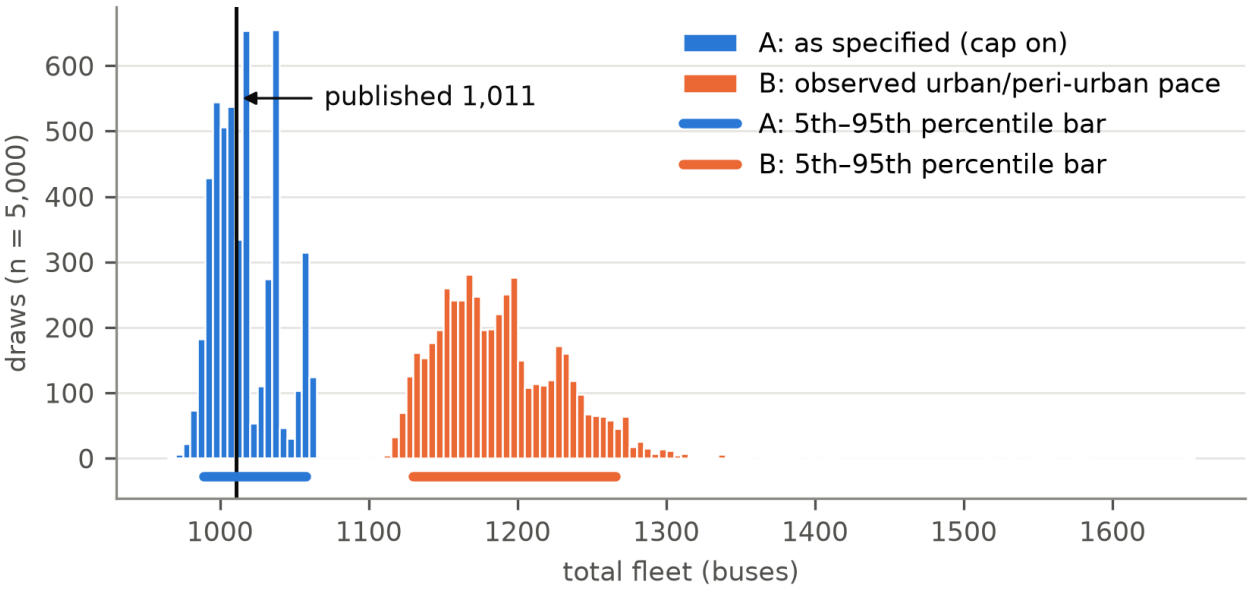


Figure 9. Fleet under joint parameter uncertainty, regimes A (as specified) and B (observed urban and peri-urban pace), against the published 1,011 [CL-56].

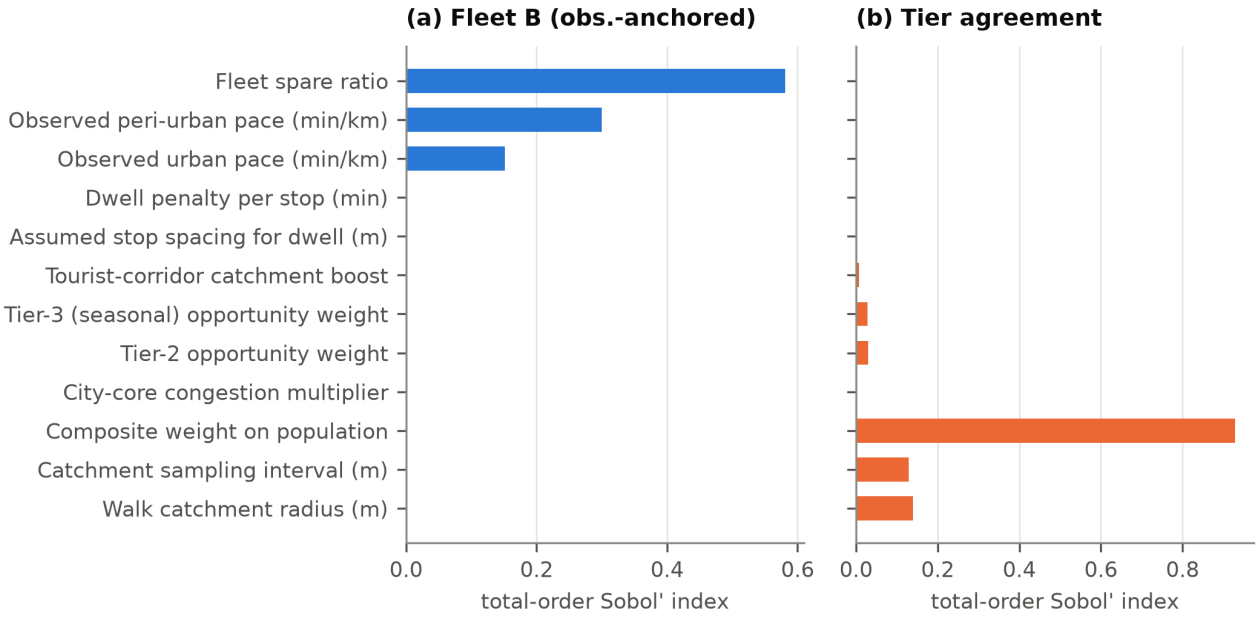


Figure 9b. Total-order Sobol' indices: which parameters drive fleet and tier uncertainty [CL-58].

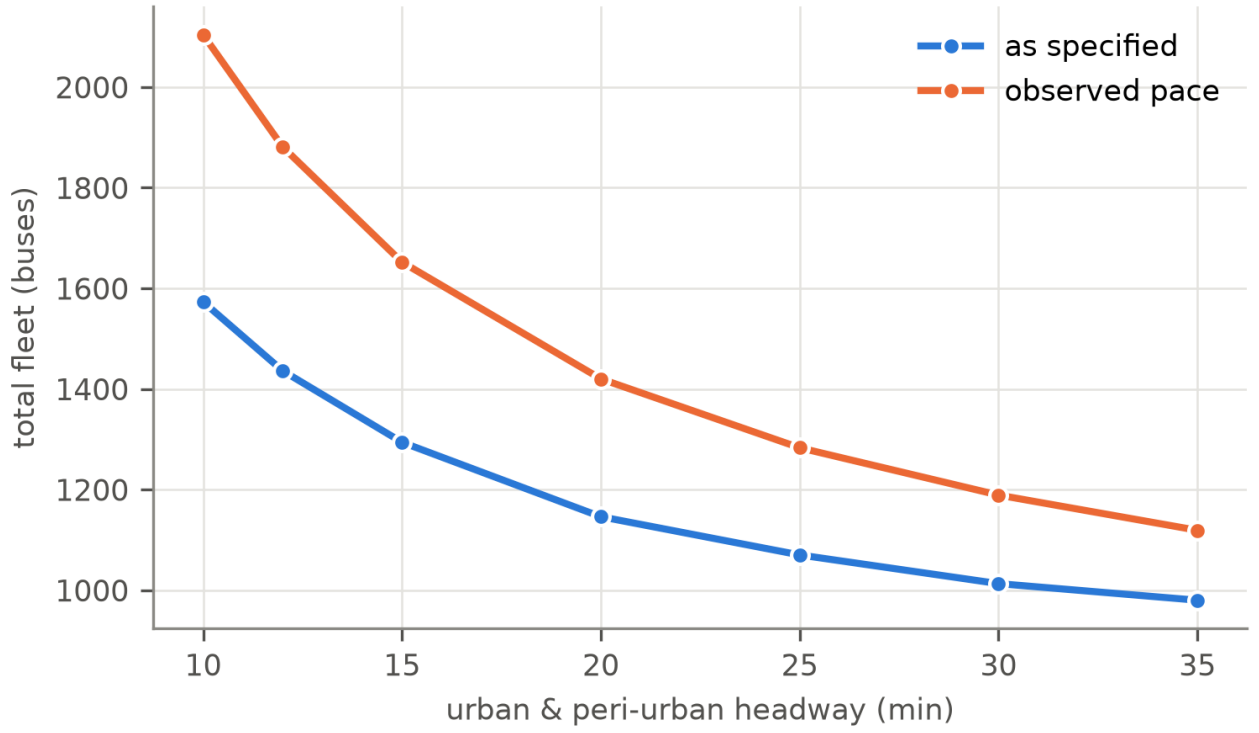


Figure S1. The fleet price of city frequency: total fleet against a common urban/peri-urban headway, as specified and at observed pace [CL-52].

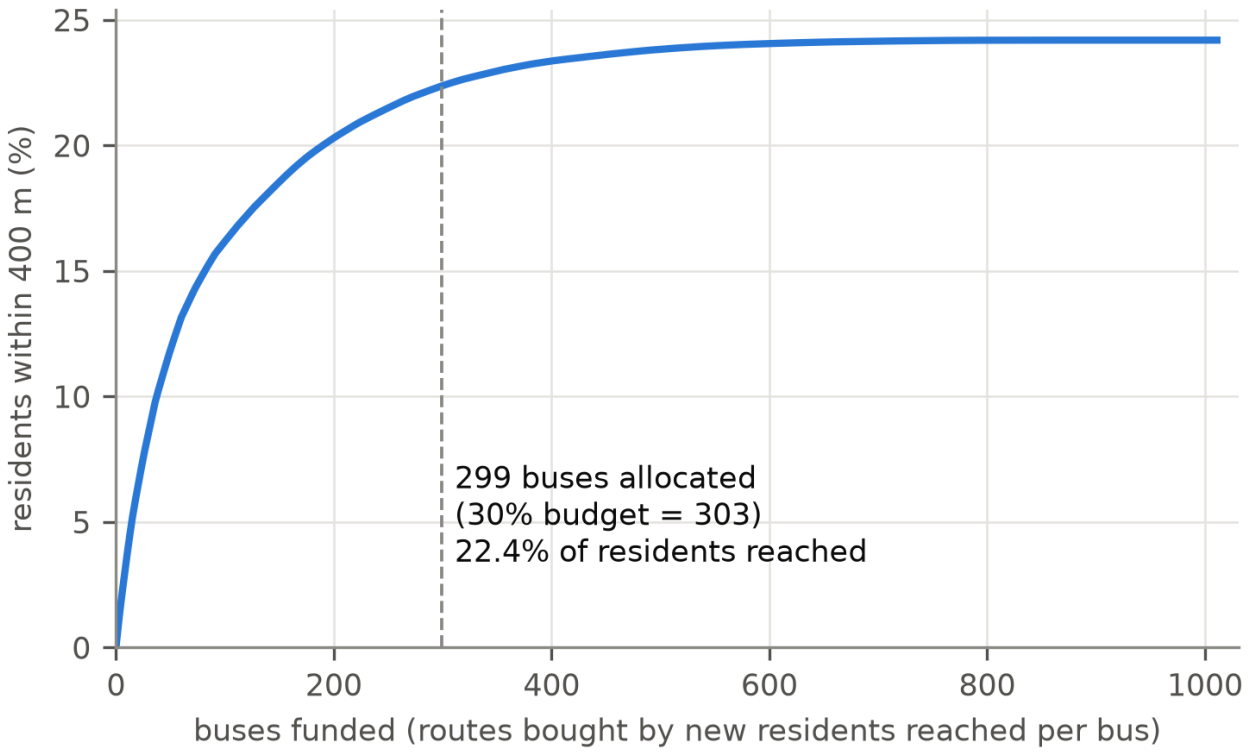


Figure S2. Funding-constrained sequencing: residents reached as routes are bought in order of new residents per bus. 30 % of the fleet reaches 92 % of the plan's coverage [CL-61].

Provenance of every number in this draft

Every quantity carries a claim ID (`CL-xx`) or a finding ID (`F1-F12`) that resolves to a row in `paper/CLAIM_LEDGER.md` and, through it, to the analysis module that produced it. The rule the repository enforces on itself is short: **a number whose module has not run does not exist and is not written into the paper.** Where a result is specified in the design but not yet computed, the draft says so in a callout instead of estimating it. That is why several subsections below name a quantity and then decline to report it.

The legacy framing that appears in older shared reports — 342 permits, 207 routes, a 39 % reduction, 95.7 % coverage, a ~1,009 fleet, and any Srinagar-metropolitan denominator — is superseded and barred from this manuscript. If any of those numbers appears in a draft you hold, that draft predates the current baseline.